Space Challenges Program 2026

Technical Design Document

Cubli

A Three-Axis Reaction-Wheel Balancing Cube

Project Team

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1 Introduction

1.1 Background & Problem Statement

The Cubli is a reaction-wheel inverted pendulum: a cube-shaped platform that balances on an edge or a corner using only internal actuation, with no external supports or contact forces beyond the single line or point of contact with the ground. The concept was introduced by the Institute for Dynamic Systems and Control (IDSC) at ETH Zurich [1, 2], whose original prototype (15 × 15 × 15 cm) mounts one brushless DC motor and flywheel on each of three orthogonal faces. Commanding a change in wheel speed produces, by conservation of angular momentum, an equal and opposite reaction torque on the cube body; braking a wheel abruptly transfers its stored angular momentum to the body in a near-impulsive event, which is what allows the cube to *jump up* from resting on a face into a balance on an edge or corner.

Dynamically, the system is an *underactuated*, nonlinear, unstable-equilibrium control problem: three actuators (the wheels) must stabilize a body with more degrees of freedom than actuated inputs, about equilibria (edge and corner balance) that are open-loop unstable and increasingly coupled across axes as the contact geometry shrinks from a line (edge) to a point (corner). This is structurally the same problem as the classical inverted pendulum, extended from a single rotational axis to a fully three-dimensional, momentum-exchange-actuated body — which is why the Cubli has become a standard benchmark platform in the controls community over the past decade.

The problem this project addresses is therefore twofold: (i) derive and validate a dynamic model and controller capable of stabilizing this underactuated system about its unstable equilibria and driving it through commanded reorientations, and (ii) realize this in hardware as a modular, largely parametric design, since final actuator and sensor specifications are not yet confirmed at the time of writing. The target capability set, in order of increasing difficulty, is edge balance, corner balance, controlled rotation, and walking (Section 1.3).

Why this is a hard build, not a decorative demo. It is worth stating plainly why a machine that looks like a cube with three flywheels resists casual replication. The difficulty is not that any single subsystem is exotic — none is — but that every subsystem sits close to a physical bound, and the bounds are *coupled*: relieving one tightens another. The following failure modes recur across published and amateur replications, and each is a consequence of that coupling rather than of poor workmanship.

- **The jump-up is an energy problem with no margin.** The wheel must store, at its maximum speed, enough angular momentum to lift the cube's center of mass over the corner. The required per-wheel inertia scales as $ML^{3/2}$ (Section 4.3.1), so mass added *anywhere* — battery, frame, fasteners — demands a larger wheel, and the wheel is itself part of M. The sizing problem is therefore a fixed-point iteration rather than a one-pass calculation, and it does not always converge: for the present cube the three-wheel set alone accounts for roughly a quarter of the total mass budget, and shrinking the wheel diameter to save space makes the situation worse, not better, because inertia must then be recovered with ballast at a smaller radius.
- **Braking torque fights the structure.** The momentum accumulated over seconds of spin-up must be shed in tens of milliseconds. That impulse passes through one motor shaft, one hub, and one motor mount — the very parts the mass budget wants thin. Many builds fail here first, and they fail in a characteristic way: the electronics are fine, the controller is sound, and the coupling, hub or mount yields on the first *successful* jump. The structural load case is set by the maneuver that proves the machine works.

- **Sensing, not the control law, sets the balancing limit.** The attitude estimate degrades from two directions at once: MEMS gyroscope bias drifts, and the accelerometer — the only absolute reference for tilt — is corrupted by vibration that the controller itself generates by spinning the wheels. Structural design and state estimation are consequently not separable concerns. A wheel half a gram out of balance at rim radius can render a perfectly correct estimator unusable, which is why wheel balance is treated here as an estimation requirement and not merely as a mechanical nicety.
- **The plant is genuinely nonlinear and constrained.** Corner balancing does not decouple into three independent single-axis problems: the axes are kinematically coupled through the contact point and the body inertia tensor. The actuators saturate in *both* torque and speed, and under any persistent disturbance the wheels drift toward their speed limit, so stored momentum must be actively managed rather than left to accumulate. Linearizing about the upright equilibrium is entirely adequate for a balancing demonstration and inadequate for anything more interesting [3].

The common thread is worth making explicit, because it determines how this project is sequenced. Replications typically fail not because the control law was wrong, but because the machine was never capable of the maneuver being asked of it: insufficient wheel inertia, motors that cannot reach their rated speed under load, a brake too slow to approximate an impulse, or a mass budget that quietly grew past the point where the stored energy suffices. Every one of these is decidable on paper, before any material is cut — and commonly is not decided there. That is the reason sizing is treated in this document as the first engineering activity rather than a detail to be settled once parts arrive.

1.2 Purpose, Application & Mission Context

The Cubli is not presented here as a balancing novelty but as a terrestrial, gravity-loaded demonstrator of **momentum-exchange attitude control** — the same physical mechanism that governs how real spacecraft point themselves. Nearly every three-axis-stabilized satellite holds and changes its attitude using reaction wheels: flywheels whose commanded speed change produces a reaction torque on the spacecraft body, exactly as in the Cubli. Spacecraft typically carry three or four such wheels (often in a pyramidal arrangement for redundancy) — directly analogous to the Cubli's three orthogonal wheels — and this reaction-wheel approach is the standard for CubeSats and small satellites in particular, making the Cubli representative of a real, currently-flown hardware class rather than an idealized toy problem.

Framed as an Attitude Determination and Control Systm (ADCS) problem, the Cubli's core behaviors map directly onto a spacecraft's day-to-day pointing task:

- **Edge / corner balance** → regulation to a commanded reference attitude and disturbance rejection — holding a pointing setpoint against perturbations.
- **Controlled rotation** → a slew maneuver: reference tracking to a new commanded attitude, exactly as a satellite re-aims a payload or antenna.
- **Jump / walk** → momentum-exchange hopping mobility, flight-proven on small bodies where wheeled traction is impossible. JAXA's Hayabusa2 mission deployed the MINERVA-II1 rovers [5] and the MASCOT lander [6] on asteroid Ryugu (2018), both of which move by spinning up and braking an internal flywheel or eccentric mass to hop across the surface, with MASCOT additionally self-righting the same way. Walking, as a sequence of controlled, directional hops, is a genuine step beyond a single ballistic hop and is an active research frontier (e.g. ETH's SpaceHopper [7]).

This gives the project two complementary application narratives rather than one: it is simultaneously a low-cost, hardware-in-the-loop testbed for standard spacecraft *attitude determination and control*, and a ground analog for the momentum-exchange *surface mobility* now used on small-body missions. Framed against a companion star-tracker project, which answers *"where am I pointing?"* (attitude determination), the Cubli answers *"drive me to that reference and hold it"* (attitude control) — the two halves of a complete ADCS loop.

Why the problem is still open. The nominal Cubli problem — balance on a corner, jump up — is solved, and was solved a decade ago on well-instrumented, precisely manufactured, generously funded hardware [1, 2, 3]. Reproducing that result is an engineering exercise, not a research contribution, and this document does not claim otherwise. The interesting questions begin immediately *after* that point, and they are not Cubli-specific: they are the questions a spacecraft attitude-control engineer faces, in a form small enough to be answered on a bench.

- **Control under simultaneous torque and momentum saturation.** A jump saturates the wheels by construction — the maneuver is defined by running them to their speed limit and dumping the momentum. Steering a nonlinear attitude system to a target while respecting a finite momentum envelope, and desaturating without losing the pointing objective, is still handled largely case-by-case with tuned heuristics rather than by a general method.
- **Recovery from large attitude excursions.** Linear controllers valid near upright say nothing useful once the body is far from equilibrium, which is precisely the regime a tumbling or mis-deployed spacecraft occupies. Nonlinear and predictive methods make claims there, and verifying those claims empirically requires a system in which a failed recovery costs a re-print rather than a mission.
- **Robustness to badly known parameters.** Real inertia tensors, friction coefficients and center-of-mass offsets are known to a few percent at best, and shift with assembly, thermal state and wear. Adaptive and robust designs are overwhelmingly validated against *simulated* parameter errors, which are drawn from the same distribution the designer assumed.
- **Whether high-performance methods survive commodity sensing.** Attitude-control techniques developed against clean measurements must eventually run on a MEMS IMU with drifting bias and vibration-corrupted accelerometers. Whether a given method degrades gracefully or collapses under that sensing is an empirical question, and it is settled by hardware rather than by analysis.

Why a Cubli, and why built from scrap. The Cubli is the cheapest honest test of the four questions above. Simulation cannot answer them, because the effects that decide the outcome are exactly the ones a model omits: unmodeled friction, frame resonance excited by the wheels, and the interaction between wheel imbalance and estimator bias. These are what separate a controller that works in simulation from one that works on hardware. At the other extreme, a full-scale testbed — air bearing, vacuum chamber, flight-grade wheels — costs orders of magnitude more and, decisively, *cannot be crashed repeatedly*. The scrap-built cube occupies the useful middle: physical enough to be honest, cheap enough to be abused. The choice to build it from commodity and salvaged parts is therefore methodological rather than merely budgetary, on four counts:

- **Cheap failure means the aggressive experiments actually get run.** A controller that must never fail cannot be tested at its limits, and a testbed too valuable to break silently biases the experimental program toward maneuvers already known to work.
- **Cheap parts are a stronger test than good parts.** Loose tolerances, an imperfectly balanced wheel and a drifting IMU are not defects to be apologized for — they *are* the parameter uncertainty and disturbance content that robust and adaptive methods claim

to tolerate. A demonstration on deliberately imperfect hardware therefore carries evidence that a demonstration on precision hardware does not.

- **Constraints force honest sizing.** With a fixed motor, a fixed print volume and a fixed budget, a marginal design cannot be rescued by specifying a larger component. This is the entire justification for the bolt-in-pocket ballast scheme of Section 5.3.4: it buys inertia at the outer radius using standard M6 hex nuts and no machined part, and whether it closes is a calculation that must be completed *before* printing, not discovered afterwards.
- **A commodity bill of materials is reproducible.** Every part is orderable or salvageable by any university group, so the result can be checked by others — which is the difference between a demonstration and a claim.

Mission context. The subsystem exercised here is the one a spacecraft uses for pointing: momentum-exchange attitude control under actuator saturation, with desaturation, on an underactuated body whose inertia is imperfectly known. In one respect the terrestrial case is strictly *harder* than the orbital one. In orbit, a marginal attitude controller degrades slowly and often invisibly — pointing error grows, wheels creep toward saturation, and the fault may take orbits to become obvious. Under gravity, the destabilizing torque never vanishes and never relents: an inadequate controller falls over immediately and unambiguously. The demonstration is, in that sense, self-evaluating, and this is a feature of the platform rather than a limitation of it.

1.3 Objectives

The project targets four behaviors, deliberately ordered as a difficulty ramp in which each milestone is a control precondition for the next:

1. **Edge balance** — stabilize the cube on a single edge, the least unstable equilibrium (contact is a line, one primary axis of instability).
2. **Corner balance** — stabilize the cube on a single vertex, the fully unstable equilibrium (point contact, all three axes coupled and unstable simultaneously).
3. **Controlled rotation** — execute a commanded slew between equilibria (e.g. edge to edge, or in-place reorientation), demonstrating reference tracking rather than pure regulation.
4. **Walking** — chain repeated, directional controlled falls between edges/corners into a net translation across a surface, re-stabilizing after each transition.

These map one-to-one onto the applications of Section 1.2: (1)–(2) demonstrate pointing regulation and disturbance rejection of increasing difficulty; (3) demonstrates slew/reference-tracking; (4) demonstrates momentum-exchange mobility of the kind flown on Hayabusa2. Milestones (1)–(3) are treated as committed deliverables for this design; walking (4) is scoped as the stretch objective, contingent on hardware confirmation and the jump-up feasibility check (angular momentum required to tip the cube vs. available motor/wheel torque), to be resolved once final components are known.

2 Project Organization

2.1 Team Organization

The project is developed by a multidisciplinary team, with responsibilities distributed across the main technical domains of the Cubli: mathematical modeling, control, mechanical/CAD design, and system integration. Table 1 summarizes each member's primary role, and the detailed responsibilities are outlined below.

Table 1: Team members, contact details and primary roles.

Member	**Contact**	**Primary Role**
Pablo Urioste Alarcón	pb.urioste4@gmail.com +34 654 491 174	System integration and lead: digital actuator and sensor coding, embedded software, and integration of the overall design.
Deyan Nikolaev Vlaev	vlaev.dido@gmail.com +359 877 082 499	Mathematical demonstration and derivation of the system dynamics.
Niccolo	—	Control system design and analysis.
Andrea Liang	andrea.liang.2006@gmail.com +31 06 5796 2976	Support on the control problem and structural/mechanical integration.
Suvanna Viriyanti Wu	Viriyanti.wu@gmail.com +39 329 074 7550	CAD modeling and mechanical design.
Athanasia Nikolova	—	CAD modeling, prototype building/assembly, and product presentation (commercial pitch).

Responsibilities.

- **Pablo Urioste Alarcón** — System integration and overall coordination. Handles the coding of the digital actuators and sensors, the embedded software stack, and the integration of the mechanical, electronic and control subsystems into a working prototype.
- **Deyan Nikolaev Vlaev** — Mathematical foundation. Responsible for the analytical demonstration and derivation of the equations of motion and the dynamic model of the Cubli.
- **Niccolo** — Control. Leads the design, tuning and analysis of the control strategy used to balance and maneuver the Cubli.
- **Andrea Liang** — Control and mechanics support. Assists on the control problem and contributes to the structural and mechanical integration of the system.
- **Suvanna Viriyanti Wu** — CAD and mechanical design. Develops the 3D models and mechanical drawings of the structure and reaction-wheel assembly.
- **Athanasia Nikolova** — CAD, prototype building and commercialization. Contributes to the 3D modeling, leads the physical building and assembly of the prototype, and is responsible for presenting and pitching the product.

2.2 Work Breakdown Structure

This plan runs from project start to the final presentation on 20 August 2026. The work is grouped into five streams — investigation, requirements and documentation; control and software; hardware design and build; three-dimensional design; and electrical and electronics — led by the owners assigned in Table 1. The investigation is kept to the essentials needed to substantiate the design: a focused literature review, the dynamics derivation, and the wheel sizing that follows from it. Table 2 breaks the streams into individual tasks, each with its scope, lead and dates.

Every task carries a single accountable lead. Where a task draws on more than one member of the team, the lead named in Table 2 is the point of contact for that task and coordinates the supporting effort, so that accountability for each item rests with one person.

Table 2: Work breakdown structure, broken into individual tasks with scope, lead and dates, grouped by workstream.

ID	Task	Scope	Lead	Dates
Investigation, Requirements & Documentation				
A1	Literature & state of the art	Review the ETH Cubli, the one-wheel Cubli, CubeSat ADCS practice and the hopping-mission precedents	Suvanna	23–25 Jul
A2	Problem statement & framing	Lock the ADCS application narrative that the rest of the document refers back to	Deyan	23–24 Jul
A3	Scope & requirements	Define scope and build the requirements table (functional, performance, design constraint) with verification methods	Deyan	24–25 Jul
A4	Milestones & roles	Prototype staging, binary gate criteria, and a one-page RACI of owners	Deyan	23–25 Jul
A5	Risk register	Trigger, impact, owner and mitigation for the leading risks	Pablo	28–29 Jul
A6	1-DoF equations of motion	Lagrangian derivation of the reaction-wheel pendulum with friction terms	Deyan	24–25 Jul
A7	3D equations of motion	Coupled three-wheel dynamics for the edge and corner pivots	Deyan	27–28 Jul
A8	Linearisation & controllability	Equilibria, linear models, controllability/observability checks	Deyan	29–31 Jul
A9	Jump-up sizing & budget	Energy-method wheel sizing and the parametric mass/inertia/CoM budget	Deyan	30 Jul–2 Aug
A10	TDD draft (all sections)	Every member drafts their section; placeholder figures acceptable	Deyan	25–28 Jul
A11	TDD deepen + rig evidence	Full modelling and control sections; insert the rig evidence and measured parameters available at the time of writing	Deyan	29 Jul–2 Aug
A12	TDD v1 assemble & submit	Consistency pass, figures, references, submission of the first assessed deliverable	Deyan	3 Aug
A13	Assessor feedback triage	Receive the assessment of the v1 submission, triage the comments and assign each to an owner	Deyan	4–7 Aug
A14	TDD v2 corrections & second draft	Address the assessor comments and incorporate the build, bring-up and test results obtained since the v1 submission	Deyan	8–13 Aug
A15	TDD v2 final submission	Final consistency pass and submission of the corrected document	Deyan	14 Aug
Control & Software				

ID	Task	Scope	Lead	Dates
B1	Sim environment + plant	Simulation setup and the 1-DoF plant, validated by energy conservation	Andrea	23 Jul–3 Aug
B2	LQR design (1-DoF, sim)	Tune the regulator and sweep robustness to inertia error, delay and saturation	Andrea	26 Jul–6 Aug
B3	State estimator	Complementary tilt filter with injected noise and bias in simulation	Nicc	27 Jul–7 Aug
B4	Firmware & comms	Firmware architecture, loop rates, and the CAN-FD command/reply layer	Nicc	24 Jul–7 Aug
B5	Sensor drivers + cal	IMU and encoder drivers, gyro/accel calibration and the lever-arm correction	Nicc	28 Jul–7 Aug
B6	Safety & mode logic	Watchdog, tilt-limit disarm, wheel-speed cap and the mode state machine	Nicc	30 Jul–7 Aug
B7	1-DoF control design complete	Consolidated 1-DoF control stack — plant, regulator, estimator and safety logic — released as the baseline for the rig	Andrea	8–9 Aug
B8	System identification	Measure K_t, wheel inertia, friction and loop latency on the rig	Andrea	7–9 Aug
B9	3D model + LQR (sim)	3D plant and the edge/corner regulators with the CAD inertia tensor	Andrea	10–14 Aug
B10	EKF / MEKF estimator	3D estimator with gyro-bias states; characterise the yaw drift	Nicc	12–15 Aug
B11	Edge balance (S2a)	Retune the edge controller on the assembled cube; log disturbance recovery	Andrea	15–17 Aug
B12	Corner balance + slew (S2b/c)	Coupled three-axis corner balance and a commanded slew; report ADCS metrics	Andrea	17–19 Aug
B13	Jump-up (S3)	Spin-up profile and brake release for a face-to-edge jump. Sacrificial scope: yields to G6 and G7 if the schedule is contested	Nicc	19–20 Aug
Hardware Design & Build				
C1	Envelope & layout study	Packaging of wheels, drivers, battery and electronics with a central CoM	Neisa	23–24 Jul
C2	Reaction-wheel design	Wheel to the inertia target: radius, rim, hub bolting to the motor bell	Suvanna	24–26 Jul
C3	Mounts & brackets	Motor mount, ring-magnet carrier and adjustable encoder bracket	Neisa	27–29 Jul
C4	1-DoF rig CAD + jigs	Single-axis rig and the balancing/assembly tooling	Suvanna	25–28 Jul
C6	Procurement order (round 1)	Order the long-lead items (rims, frame stock, connectors, spares)	Pablo	23–24 Jul
C7	Bench PSU & bring-up	Current-limited supply, safety kit, and single-axis electrical bring-up	Pablo	25–27 Jul
C8	Fabricate wheel rings	Print the PET-CF rings and hubs to the frozen wheel geometry	Suvanna	27 Jul–2 Aug
C8b	Wheel ballast & completion	Install the M6 bolt/nut ballast stations to the sizing specification and complete each wheel	Suvanna	4–6 Aug
C9	Balance & spin-test	Static-balance and containment spin-test every wheel, including a per-wheel ballast-count verification	Suvanna	5–6 Aug

ID	Task	Scope	Lead	Dates
C10	Assemble 1-DoF rig	Motor, balanced wheel, encoder gap, IMU and hard stops, on reinforced rig supports	Pablo	31 Jul–2 Aug
3D Design				
D1	Concept sizing & fit study	Confirm that every component fits the 150 mm envelope and lay out the internal structure	Neisa	27 Jul–2 Aug
D2	Frame design freeze	Freeze the inertia-bearing frame geometry. Later features that do not affect the mass properties may still be added	Neisa	2 Aug
D3	Mass properties & budget closure	Export the inertia tensor and CoM and close the mass budget against the sizing analysis	Neisa	3–4 Aug
D4	Detail CAD	Brackets, electronics bay, harness routing and the electrical keep-outs to the frozen board outline	Suvanna	3–5 Aug
D5	Procurement order (round 2)	Consolidate the bill of materials arising from the 3D design and place the second order	Pablo	5 Aug
D6	Tolerance & clearance analysis	Fit and tolerance study, including the wheel sweep envelope and fastener access	Neisa	5–6 Aug
D7	Manufacturing package release	Issue the drawings, print set and assembly instructions for fabrication	Neisa	7 Aug
C11	Print structure + frame	Structural print set and the fabricated, orthogonal frame	Suvanna	8–12 Aug
C12	Perfboard + harness	Power/signal board and the full power and CAN harness	Pablo	8–12 Aug
C13	Cube integration + power-on	Assemble the three axes, then bench and battery smoke test	Pablo	13–15 Aug
C14	Brake mechanism build	Fabricate and fit the servo brake; bench-test the deceleration	Suvanna	15–18 Aug
Electrical & Electronics				
E1	Wiring & placement design	Electrical architecture, wiring scheme and component placement for the single-axis rig, issued as the build reference	Pablo	28–31 Jul
E2	Rail, driver & encoder bring-up	5 V rail soldered and trimmed to 5.00 V under load; driver phase-wired and FOC-calibrated; MA600 encoder mounted at ≤ 0.5 mm air gap with verified read-back. Open-loop operation on the bench supply	Pablo	1–3 Aug
E3	Rig bus & sensor integration	Two-node CAN-FD bus with split termination at both physical ends, Teensy and IMU integrated on the rig harness	Pablo	4–5 Aug
E4	Board outline & floorplan	Driver placement from the encoder cable limit, CAN routing, and the perfboard floorplan issued as a frozen outline to CAD	Pablo	4–5 Aug
E5	1-DoF closed-loop bring-up	Closed loop on the rig with the disarm path and watchdog verified before any balance attempt	Pablo	6–7 Aug

ID	Task	Scope	Lead	Dates
E6	Rig parameter measurement	Instrumented measurement of K_t, wheel inertia, friction and loop latency, delivered to the control workstream	Pablo	7–9 Aug
E7	Perfboard + battery power chain	Populate the distribution perfboard (two ground domains, one star point, split 5 V rail); XT90 chain with anti-spark, fuse and ≥ 50 V bulk capacitance; replicate the remaining actuation sets	Pablo	8–11 Aug
E8	4-node bus + cube harness	Bus scaled to four nodes with stubs < 10 cm and an error soak at 5 Mbit/s; full power and signal harness fabricated and routed; Wi-Fi telemetry for untethered operation	Pablo	11–13 Aug
E9	Cube power-on, 48 A + thermal	First battery power-on and the three-axis 48 A transient; driver thermals in the enclosed volume; all axes addressable	Pablo	13–15 Aug
E10	Brake servo rail + measurement	Servo on an isolated 5 V rail with PWM drive and flyback; rail stability under stall, brake closure time and spin-down brake torque measured	Pablo	16–19 Aug
Deliverables				
F1	Results, demo & final presentation	Slow-motion capture, results plots, deck, and the final presentation	Suvanna	18–20 Aug

2.3 Schedule and Contingency Plan

The full-page schedule in Figure 1 places every task of Section 2.2 on the 23 July–20 August 2026 calendar, and Table 3 states the gate that closes each phase.

Structure of the plan. The first two weeks establish the analytical and design foundation: the dynamics derivation and wheel sizing on the documentation side, the single-axis rig and its wiring design on the hardware and electrical sides, and the simulation environment on the control side. These converge on 2 August, which carries two objectives: the first powered bring-up of the single-axis rig, taking the 5 V rail, the driver and the encoder to verified open-loop operation (M1); and the freeze of the three-dimensional frame geometry, which closes the mass and inertia budget against the sizing analysis (M2). The frame freeze releases the second procurement round on 5 August (M3), which is the last point at which parts can be ordered with useful delivery margin before cube assembly begins on 13 August.

The documentation stream carries two assessed submissions. The first is delivered on 3 August (G4) with the evidence available at that date; the assessor's comments are triaged during the following week and the corrected second draft is submitted on 14 August (M8), by which point the rig results and the integrated cube can both be reported.

Critical path and contingency. The critical path runs from the 2 August bring-up through the closed-loop rig (M4), the measured plant parameters (M6) and the single-axis control design (M7), into the three-dimensional controller and the balancing demonstrations. The single-axis rig is the protected item, since its balance result is the experimental evidence in the document. The three-dimensional design proceeds in parallel on the mechanical side, so that fabrication is not held by the control work.

The plan carries the full scope through to the jump-up demonstration. The final week is nonetheless tightly constrained: edge balance begins two days after cube assembly completes, and the jump-up gate falls on the day of the final presentation. The jump-up (S3) is therefore designated sacrificial scope by standing decision. If the balancing demonstrations require the time, the jump-up yields first, and the demonstration presented on 20 August is the corner balance and commanded slew. The brake mechanism build and its associated electrical work are sequenced after the edge-balance gate for the same reason.

Table 3: Milestone gates, from kick-off to the final presentation, with the criterion that closes each one.

Gate	Pass criterion	Date
G1	Long-lead order placed; requirements and plan locked	24 Jul
G2	Dynamics, linearisation and wheel sizing complete	31 Jul
M1	1-DoF rig bring-up: 5 V rail trimmed under load, driver commutating, encoder verified at ≤ 0.5 mm gap	2 Aug
M2	3D frame geometry frozen at 150 mm; mass, inertia and CoM budget closed against the sizing analysis	2 Aug
G4	TDD v1 delivered for assessment	3 Aug
M3	Procurement round 2 consolidated and ordered	5 Aug
M4	1-DoF closed loop live; disarm path and watchdog verified	7 Aug
M5	Manufacturing package released for fabrication	7 Aug
M6	Plant parameters measured on the rig and validated against simulation	9 Aug
M7	1-DoF control design complete: plant, regulator, estimator and safety logic	9 Aug
G3	1-DoF rig balances $\geq$60 s unassisted	12 Aug
M8	TDD v2 corrected and finally submitted	14 Aug
G5	Cube integrated and powered; all axes addressable	15 Aug
G6	Edge balance (S2a) for $\geq$60 s with disturbance recovery	17 Aug
G7	Corner balance and commanded slew (S2b/c)	19 Aug
G8	Face-to-edge jump-up (S3) captured on video (sacrificial scope)	20 Aug
G9	Final presentation and demonstration	20 Aug

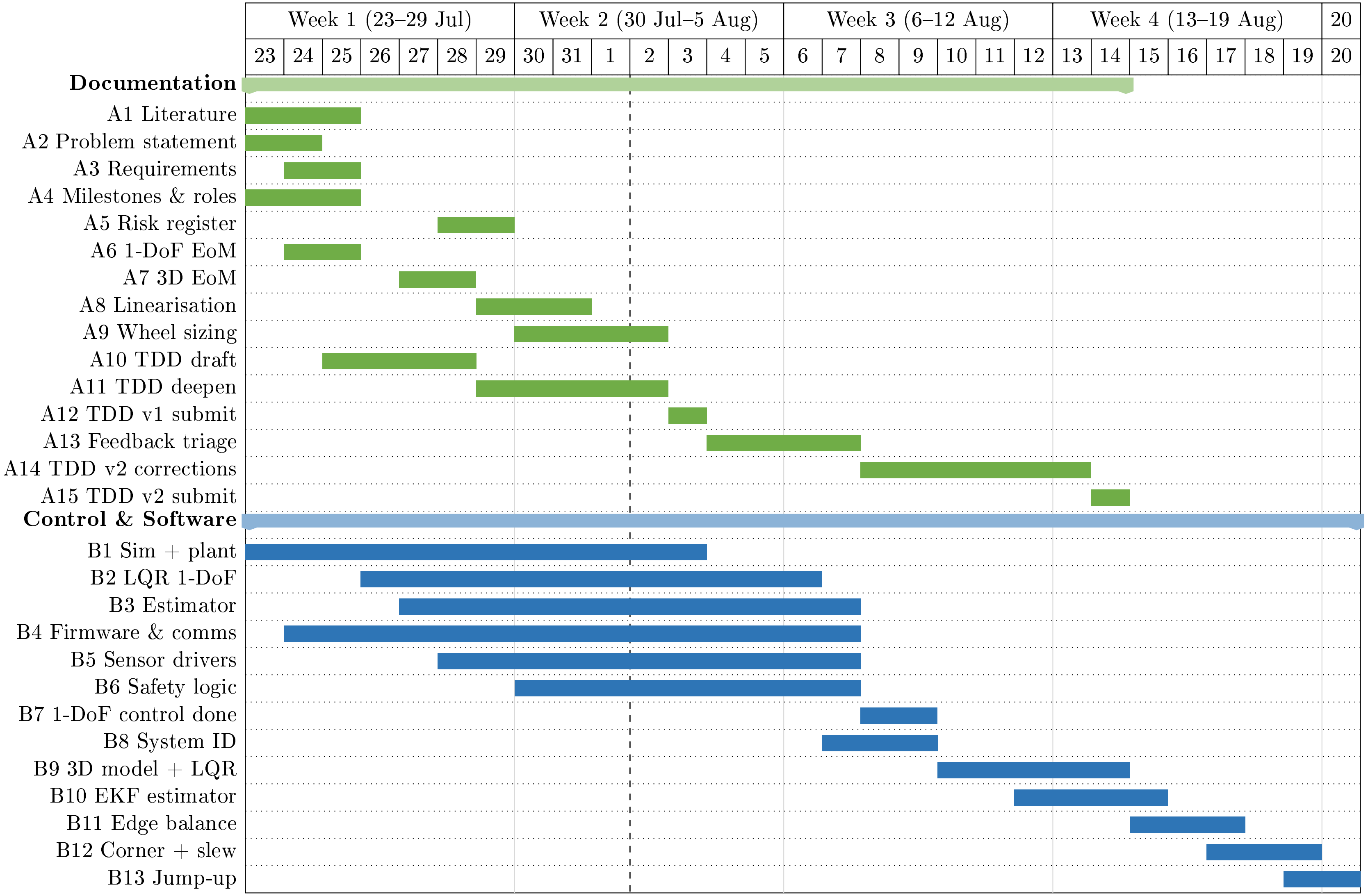

Figure 1: Project schedule, 23 July–20 August 2026 (sheet 1 of 3): documentation and control/software streams. The documentation stream (green) carries two assessed submissions, on 3 and 14 August. Control and software (blue) runs from the simulation environment through to the balancing demonstrations, with the three-dimensional extension concentrated in the final two weeks. The dashed vertical rule marks the current date at the time of compilation. The build streams continue in Figure 2 and the milestone gates in Figure 3.

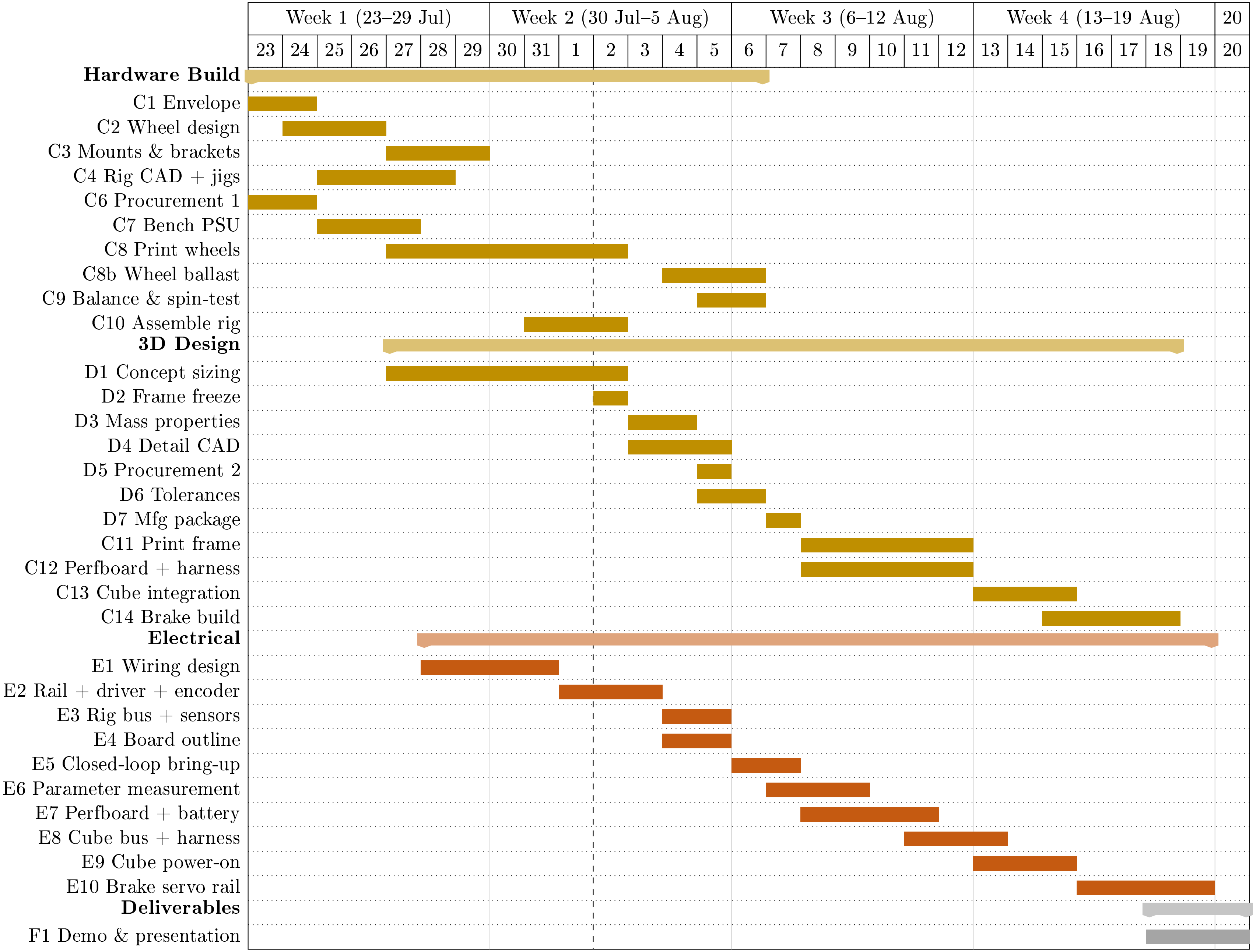

Figure 2: Project schedule, 23 July–20 August 2026 (sheet 2 of 3): hardware, three-dimensional design, electrical and deliverable streams, continued from Figure 1. Hardware (gold) separates the single-axis build from the three-dimensional design, and the electrical stream (orange) proceeds from the single-axis

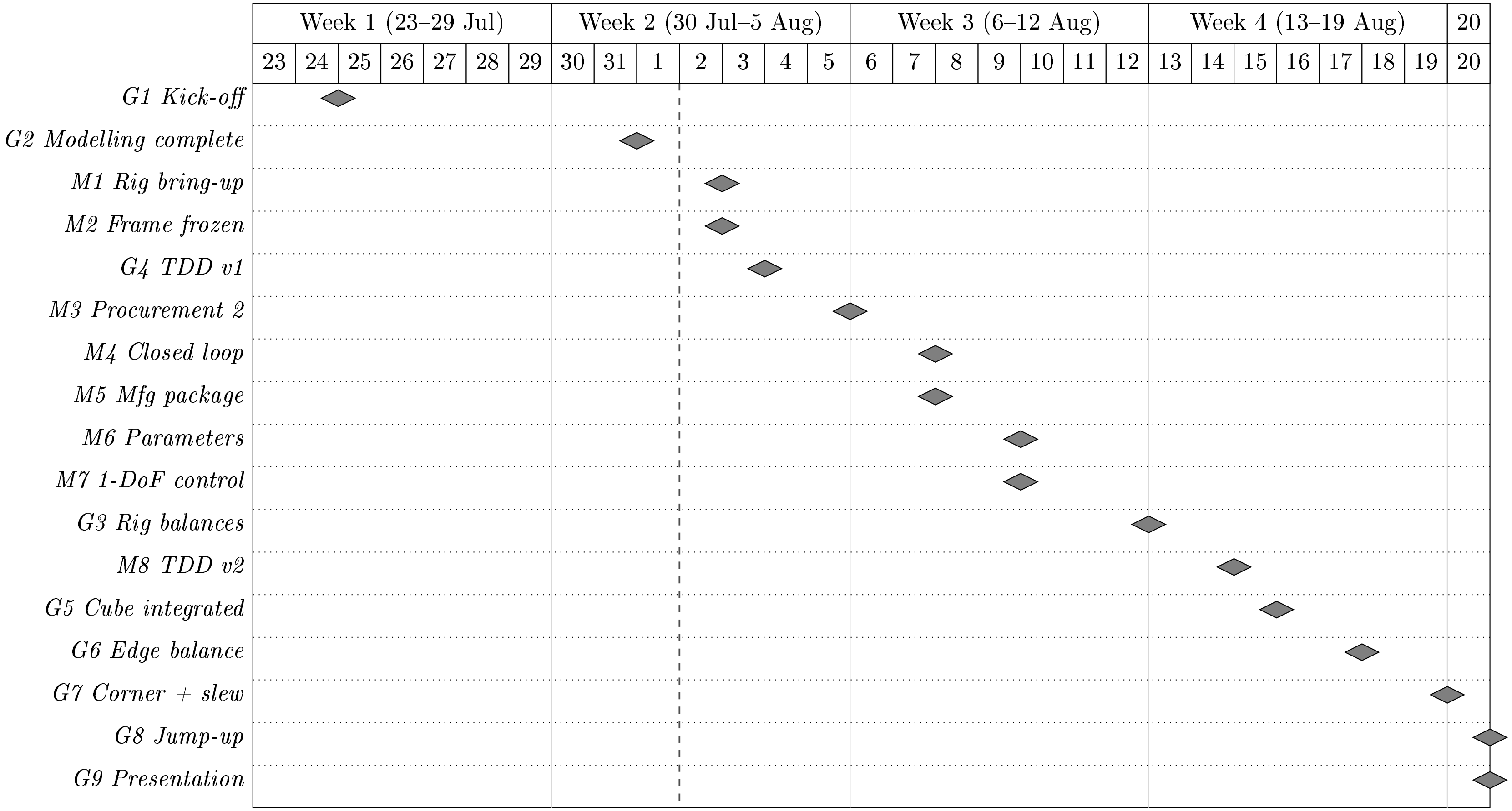

Figure 3: Project schedule, 23 July–20 August 2026 (sheet 3 of 3): the milestone gates that close each phase, continued from Figures 1 and 2. Grey diamonds mark the gates listed in Table 3, placed on the day each gate falls due. The dashed vertical rule marks the current date at the time of compilation. All three sheets share the same time axis, so columns align between them.

3 System Analysis & Requirements

3.1 Scope of Work & Use Cases

3.2 Actuation Trade-off — Why Reaction Wheels

Since this is a proof-of-concept demonstration of the control loop, a system actuated with orthogonal reaction wheels is well-suited to represent principles applicable to smallsat ADCS. The momentum-exchange actuation principle and closed-loop control architecture are the same as those used on real spacecraft, even though the Cubli's dynamics (balancing against gravity) differ from a satellite's free-fall attitude control problem. Reaction wheels are also structurally simpler and require less specialized hardware than control moment gyros, which offer higher torque density but at the cost of gimbal mechanisms and more complex control software, complexity that is typically only justified on larger spacecraft. For a low-cost, resource-limited project, a reaction wheel Cubli therefore demonstrates the core actuation and control principles of smallsat ADCS in a practical and relevant way.

3.3 Requirements & Objectives Traceability

Each project objective (Section 1.3) imposes a required capability, which in turn is confirmed by a specific test in the validation plan (Section 6). The mapping below keeps every requirement traceable from motivation to verification.

Table 4: Objective-to-requirement-to-verification traceability.

Objective	Required capability	Verified by
Edge balance	Regulation about a line-contact equilibrium	Sec. 6.9.1
Corner balance	Regulation about a point-contact equilibrium, all axes	Sec. 6.9.2
Controlled rotation	Reference tracking / slew between equilibria	Sec. 6.9.2
Walking (stretch)	Chained jump-up + re-stabilization	Sec. 6.9.3

3.4 Coordinate Frames & Conventions

4 Dynamic Modeling & Control

The full Cubli is underactuated and simultaneously unstable about all three axes when balanced on a corner (Section 1.3). The modeling, identification, and control workflow is therefore first validated on a two-dimensional (2D) single-face prototype with one rotational degree of freedom, before being generalized to the 3D cube in Section 4.5.

4.1 2D Reaction-Wheel Pendulum

The 2D prototype is a 3D-printed plate sized to one cube face, with a motor-driven momentum wheel at its center and a braking mechanism at one corner. The plate pivots on a bearing at that corner, constraining motion to a single rotational degree of freedom in a vertical plane.

4.1.1 Equations of Motion

The nonlinear equations of motion are

$$\ddot{\theta}_b = \frac{(m_b l_b + m_w l)\, g \sin\theta_b - T_m - C_b\dot{\theta}_b + C_w\dot{\theta}_w}{I_b + m_w l^2}, \tag{1}$$

$$\ddot{\theta}_w = \frac{(I_b + I_w + m_w l^2)(T_m - C_w\dot{\theta}_w)}{I_w(I_b + m_w l^2)} - \frac{(m_b l_b + m_w l)\, g \sin\theta_b - C_b\dot{\theta}_b}{I_b + m_w l^2}, \tag{2}$$

with motor current-torque relation $T_m = K_m u$. Symbols are defined in Table 5 and used throughout this section.

Table 5: 2D pendulum model variables.

Symbol	Description	Unit
θ_b	Body tilt angle	rad
θ_w	Wheel rotation relative to body	rad
m_b, m_w	Body, wheel mass	kg
I_b, I_w	Body (about pivot), wheel (about motor axis) inertia	$\mathrm{kg\,m^2}$
l	Motor axis to pivot distance	m
l_b	Body CoM to pivot distance	m
g	Gravitational acceleration	$\mathrm{m/s^2}$
T_m	Motor torque	N m
C_b, C_w	Body, wheel damping coefficient	$\mathrm{kg\,m^2/s}$
K_m	Motor torque constant	N m/A
u	Motor current input	A

4.2 Parameter Identification & the 2D Testbed

CAD mass-properties simulation, accounting for print material, infill, and mounted hardware (driver, encoder, MCU, gyroscope/accelerometer, wiring), gives m_b, m_w, I_b, I_w directly, while l and l_b follow from the plate's design geometry. The damping coefficients C_b and C_w cannot be obtained from CAD, since they depend on bearing friction, wiring drag, and motor cogging; nominal values will be estimated for the initial control design, with dedicated testing planned once the prototype is fabricated.

Table 6: Parameter determination method (symbols per Table 5).

Symbol	Determination method
l	Calculated during design
l_b	Calculated during design
m_b	CAD mass properties
m_w	CAD mass properties
I_b	CAD mass properties
I_w	CAD mass properties
C_b	Estimated (future testing planned with prototype)
C_w	Estimated (future testing planned with prototype)
K_m	Motor datasheet

4.3 Jump-Up Feasibility Analysis

Given CAD-derived $I_b, I_w, m_b, l_b, m_w, l$, the wheel speed ω_w required for jump-up is estimated before fabrication, assuming a perfectly inelastic wheel-body collision. Conservation of angular momentum at impact and of energy from impact to edge-standing yield

$$\omega_w^2 = (2-\sqrt{2})\,\frac{I_w + I_b + m_w l^2}{I_w^2}\,(m_b l_b + m_w l)\,g. \tag{3}$$

This value is checked against the motor's achievable speed at the available bus voltage and used to size wheel inertia (radius, rim mass) within the motor's operating range without excessively reducing the balancing recovery angle. Post-fabrication, ω_w is refined experimentally to correct for deviations from the ideal collision assumption.

The braking mechanism uses a servo-driven barrier that strikes a bolt head on the wheel to stop it abruptly. Servo response time bounds the validity of the "instantaneous stop" assumption; a slower stop yields a lower effective post-impact body velocity than predicted, compensated through the same tuning process.

4.3.1 Torque-Limited Wheel-Inertia Target

The relation above assumes an instantaneous, lossless momentum transfer. For preliminary sizing, the achievable motor speed is treated as fixed and the required per-wheel inertia is derived from a momentum budget that additionally accounts for finite brake torque. This gives an ideal per-wheel angular momentum

$$h_{w,\mathrm{ideal}} = \sqrt{\eta}\,\frac{\sqrt{2g\,\lambda\,\kappa}}{n}\,ML^{3/2}, \tag{4}$$

where M, L, g are cube mass, edge length, and gravity, η is an energy margin, and κ, λ, n are dimensionless geometric factors set by the contact case (Table 7).

The corner case is more demanding and governs the design.

Table 7: Contact-case geometric factors.

Case	κ (pivot inertia)	λ (CoM rise)	n (momentum transfer)
Edge	2/3	$1/\sqrt{2}-1/2$	1
Corner (governs design)	11/12	$\sqrt{3}/2-1/2$	$\sqrt{2}$

A finite brake torque τ_b cannot deliver the impulse instantaneously and must exceed the gravitational tip-over torque $\tau_g = MgL/2$. This loss is captured by a brake-amplification factor

$$\beta = \frac{\tau_b}{\tau_b - \tau_g}, \qquad h_w = \beta\, h_{w,\text{ideal}}, \qquad I_{w,\text{target}} = \frac{h_w}{\omega_{\max}}, \tag{5}$$

where $\omega_{\max}$ is the maximum practical wheel speed. Equations (4) and (5) are the sizing relations used at step 4 of the closure procedure of Section 5.2: given a candidate cube mass and edge length they return the inertia the wheel of Section 5.3.4 must meet. Their inputs are recorded in Table 8 and are not fixed here—M and L are outputs of the packaging closure, not assumptions preceding it.

Two features of these relations govern how the iteration behaves. First, $h_w \propto ML^{3/2}$, so growing the cube to accommodate a larger wheel also raises the requirement; convergence depends on the achievable inertia growing faster ($\sim D_w^2$) than the requirement does. Second, β diverges as $\tau_b \to \tau_g$: a brake whose torque only marginally exceeds the gravitational tip-over torque cannot deliver the required impulse at any wheel speed. The brake torque must therefore be checked against $\tau_g = MgL/2$ on every iteration, and a comfortable ratio τ_b/τ_g maintained rather than a merely positive margin.

Table 8: Torque-limited wheel-inertia target (corner case). Inputs are fixed by the closure procedure of Section 5.2; the table is populated on each iteration. τ_b is an assumed value pending measurement, justified below and equivalent to a 38 ms stop at the present h_w.

Symbol	Description	Value
M	Cube mass	TBD
L	Cube edge length	150 mm
η	Energy margin	TBD
τ_b	Brake torque per wheel	5 N m (assumed; see below)
$\tau_g = MgL/2$	Gravitational tip-over floor	TBD
β	Brake-amplification factor	TBD
$\omega_{\max}$	Max wheel speed (design)	6000 rpm
h_w	Required angular momentum	0.198
$I_{w,\text{target}}$	**Target wheel inertia**	3.15×10^{-4} kg m^2

Choice of $\omega_{\max}$. The maximum wheel speed is a design choice rather than a datasheet figure, and it enters the sizing directly through $I_{w,\text{target}} = h_w/\omega_{\max}$: a higher assumed speed buys a proportionally smaller wheel. It is therefore the natural relief valve when the envelope caps the wheel below the requirement (step 5, Table 11), but it must be set from what the drive can sustain in service, not from the no-load figure $K_v V_{\text{bus}}$. Three effects separate the two: winding resistance, which costs speed under the torque required to spin the wheel up; the reduced phase voltage available under sinusoidal field-oriented modulation; and the fall of pack voltage through the discharge curve, which must be taken at its lower end if the maneuver is to remain available on a partly-depleted pack rather than only on a fresh one. The design value is set below the ceiling those effects imply, leaving margin for motor tolerance, bearing and aerodynamic drag, and the residual uncertainty in M. The resulting back-EMF should remain a comfortable fraction of the usable bus voltage, and the electrical frequency $\omega_{\max} p/2\pi$ at p pole pairs must sit inside the driver's commutation capability—both checks to be confirmed once the motor and pack are selected.

Choice of τ_b, and what it is not. The brake torque requires more care than the other inputs, because the mechanism does not generate it in the way the symbol suggests. The brake is a servo-driven barrier struck by a bolt head on the wheel: the servo only *holds* the barrier in the wheel's path, and the wheel's own momentum does the work of the collision. The servo's stall torque is therefore *not* τ_b and need not approach it—the TowerPro MG92B of Table 26 is rated near 0.29 N m, roughly one seventeenth of the value used in sizing, and this is not an inconsistency. It must resist the barrier being deflected, not decelerate the wheel.

Because the stop is a collision rather than a friction clutch, τ_b is not an independent property of any component and cannot be read from a datasheet. It is fixed entirely by how quickly the angular momentum is destroyed,

$$\tau_b = \frac{h_w}{\Delta t}, \tag{6}$$

so quoting a brake torque is equivalent to asserting a stopping time. At the present operating point ($h_w \approx 0.19\,\mathrm{kg\,m^2/s}$) the design value $\tau_b = 5\,\mathrm{N\,m}$ corresponds to $\Delta t \approx 38\,\mathrm{ms}$. Stated that way the assumption is open to inspection: 38 ms is characteristic of servo travel rather than of a rigid impact, and a bolt head striking a printed barrier should arrest the wheel in a few milliseconds, implying a substantially larger τ_b. The design value is thus expected to be conservative for *sizing* purposes.

That conservatism is deliberate, and it is asymmetric. Since $\beta = \tau_b/(\tau_b - \tau_g)$ falls towards unity as τ_b grows, underestimating the brake torque inflates the inertia requirement—the safe direction—while overestimating it leaves the wheel undersized. With the present τ_g, raising τ_b from 5–10 N m relaxes $I_{w,\text{target}}$ by only about 12 %, whereas lowering it to 2 N m inflates the requirement by roughly 65 % and to 1.5 N m by over 150 %. The penalty for being wrong is concentrated almost entirely on the low side, which is why a mid-range value is adopted in preference to the higher figure a short impact would justify.

One consequence must be carried into the structural design rather than left implicit: τ_b is also a *load*. The sizing value is chosen for the dynamics, and the same number should not be used to size the barrier, the bolt, the servo bracket or the wheel spokes, since a genuine few-millisecond stop would put an order of magnitude more torque through that load path. Until measurement settles the question, the structure is to be checked against a short-Δt case and the dynamics against $\tau_b = 5\,\mathrm{N\,m}$. The measurement that closes this is the spin-down test of Section 6.9.3, which should report both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$, which is the sizing quantity, and the peak torque, which is the structural one; the two may differ several-fold, and the logging rate must be fast enough to resolve an event of a few milliseconds without smoothing the peak away.

4.4 Control Design on the 2D Model

4.4.1 LQR Baseline Control

Linearizing about the upright equilibrium $(\theta_b, \dot{\theta}_b, \dot{\theta}_w) = (0, 0, 0)$ gives

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta_b, \dot{\theta}_b, \dot{\theta}_w), \tag{7}$$

with A, B built from the parameters in Table 5. Zero-order-hold discretization at a rate set by the open-loop unstable pole, with a safety margin, gives $x[k+1] = A_d x[k] + B_d u[k]$. A discrete LQR gain $K_d = (K_{d1}, K_{d2}, K_{d3})$ minimizes

$$J(u) = \sum_k x^{\mathrm{T}}[k]Qx[k] + u^{\mathrm{T}}[k]Ru[k], \tag{8}$$

giving $u[k] = -K_d(\hat{\theta}_b[k], \hat{\dot{\theta}}_b[k], \hat{\dot{\theta}}_w[k])$.

The tilt estimate $\hat{\theta}_b$ is obtained from a gyroscope and accelerometer mounted at, or as close as mechanically feasible to, the pivot point rather than at the plate's center. This placement is necessary because an accelerometer at radius r from the pivot measures gravity plus the tangential ($r\ddot{\theta}_b$) and centripetal ($r\dot{\theta}_b^2$) acceleration components induced by the plate's rotation, which corrupt the naive estimate $\tan^{-1}(-a_x/a_y)$ during fast motion (e.g., jump-up or aggressive recovery); placing the sensor at $r \approx 0$ makes both terms negligible. Tilt is estimated with a complementary filter combining gyro integration (drift-prone, unaffected by the terms above) with accelerometer inclination (drift-free, but unreliable during shocks):

$$\hat{\theta}_b[k] = (1-\gamma)\big(\hat{\theta}_b[k-1] + \dot{\theta}_{b,\mathrm{gyro}}[k]\,\Delta t\big) + \gamma\,\theta_{b,\mathrm{accel}}[k], \tag{9}$$

where $\theta_{b,\mathrm{accel}}[k] = \tan^{-1}(-a_x[k]/a_y[k])$ is the instantaneous accelerometer-only estimate and $\gamma \in (0,1)$ is a small weight favoring the gyro term between corrections.

A constant tilt-sensor offset d, including residual bias from imperfect pivot placement or sensor misalignment, would otherwise appear as a nonzero steady-state wheel speed rather than a corrected angle. A low-pass filter state

$$x_f[k+1] = (1-\alpha)x_f[k] + \alpha\hat{\theta}_b[k] \tag{10}$$

is subtracted from the tilt feedback,

$$u[k] = -K_d(\hat{\theta}_b[k] - x_f[k],\ \hat{\dot{\theta}}_b[k],\ \hat{\dot{\theta}}_w[k]), \tag{11}$$

so d is absorbed by x_f at steady state, requiring no separate calibration stage. This fixed-gain design is the baseline balancing controller; Table 9 summarizes the control variables.

4.4.2 Gain Scheduling & Mode Transitions

The controller switches from the open-loop jump-up sequence to the LQR law once $|\hat{\theta}_b| < \theta_{\mathrm{th}}$, with hysteresis around θ_{th} to prevent mode chattering. The baseline uses a single fixed gain K_d at the upright equilibrium. If recovery degrades near the edges of the operating range, where the small-angle linearization weakens, gain scheduling will be added: additional gains $K_d^{(i)}$ computed at several linearization points, with the controller interpolating between them based on $\hat{\theta}_b$, reusing the same discrete structure and offset filter.

4.5 Generalization to the 3D Cube

The dynamics, identification workflow, and control approach validated on the 2D model extend to the full three-dimensional cube, where three orthogonal reaction wheels must stabilize the body about its edge and corner equilibria.

4.5.1 Full-Body Equations of Motion

With $R \in SO(3)$ the cube's orientation and $\omega_b \in \mathbb{R}^3$ its body-frame angular velocity, the attitude kinematics are

$$\dot{R} = R\,[\omega_b]_\times, \tag{12}$$

Table 9: Control-design variables (2D and corner-balancing 3D model).

Symbol	Description
x	State vector
A, B	Continuous-time state/input matrices (linearized model)
A_d, B_d	Discrete-time (ZOH) counterparts of A, B
K_d	LQR feedback gain
Q, R	LQR state/input weighting matrices
$J(u)$	LQR cost function
$\hat{\theta}_b, \hat{\dot{\theta}}_b$	Estimated tilt angle, rate (gyroscope/accelerometer)
$\hat{\dot{\theta}}_w$	Estimated wheel rate (encoder)
a_x, a_y	Raw accelerometer readings (tangential, radial axes)
$\theta_{b,\text{accel}}$	Instantaneous accelerometer-only tilt estimate
$\dot{\theta}_{b,\text{gyro}}$	Raw gyro rate measurement
γ	Complementary-filter weight (accelerometer term)
Δt	Control/estimation loop sample period
x_f, α	Offset-correction filter state, filter coefficient
d	Constant tilt-sensor offset
θ_{th}	Tilt threshold, jump-up $\rightarrow$ balancing switch
$K_d^{(i)}$	Scheduled gain at linearization point i (extension)

where $[\cdot]_\times$ is the skew-symmetric cross-product matrix. The assembly's angular momentum about the pivot, expressed in the body frame, generalizes the single-axis momentum term of the 2D model:

$$H = I\omega_b + \sum_{i=1}^{3} I_{w,i}\, \omega_{w,i}\, e_i. \tag{13}$$

Euler's equation gives

$$\dot{H} + \omega_b \times H = \tau_g - C_b\, \omega_b, \tag{14}$$

and each wheel obeys

$$I_{w,i}\big(\dot{\omega}_{b,i} + \dot{\omega}_{w,i}\big) = T_{m,i} - C_{w,i}\, \omega_{w,i}, \qquad i = 1, 2, 3. \tag{15}$$

Table 10 defines all symbols.

Restricting motion to a single vertical plane with two wheel torques set to zero recovers the 2D prototype model exactly, with $I_b, l_b, m_b, m_w, I_w, C_b, C_w$ the corresponding single-axis projections of $I, r_{cm}, m, I_{w,i}, C_{w,i}$.

4.5.2 Edge vs. Corner Equilibria & Linearization

Jump-up proceeds in two stages: Flat-to-Edge, where one wheel brakes, followed by Edge-to-Corner, where a second wheel brakes. **Edge balancing** constrains the body to a single axis, so the dynamics reduce to the 2D form, and the 2D LQR baseline and offset filter apply directly, with m_b, l_b, I_b replaced by their edge equivalents.

Corner balancing is genuinely 3D: all three wheels contribute across three rotational DOF. Linearizing about the corner-up equilibrium $(\theta, \omega_b, \omega_w) = (0, 0, 0)$, with $\theta \in \mathbb{R}^3$ the small orientation error, gives the same state-space form as the 2D case:

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad x := (\theta, \omega_b, \omega_w) \in \mathbb{R}^9, \tag{16}$$

Table 10: Full-body (3D) model variables.

Symbol	Description
R	Cube orientation relative to world frame
ω_b	Body angular velocity, body frame ($\in \mathbb{R}^3$)
$\omega_{b,i} = e_i^{\mathrm{T}} \omega_b$	Body angular velocity about wheel axis i
$\omega_w = (\omega_{w,1}, \omega_{w,2}, \omega_{w,3})$	Wheel angular velocities relative to body
e_i	Spin axis of wheel i (face normal)
I	3×3 inertia tensor, body+wheels, about pivot
H	Total angular momentum about pivot
$\tau_g = m\, r_{cm} \times (R^{\mathrm{T}} \hat{g})$	Gravity torque about pivot
m	Total assembly mass
r_{cm}	Body-frame vector, pivot to assembly CoM
$\hat{g}$	World vertical unit vector
C_b	3×3 body damping matrix
$I_{w,i}, C_{w,i}$	Inertia, damping of wheel i
$T_{m,i} = K_{m,i} u_i$	Motor torque, wheel i
$u_i, K_{m,i}$	Motor current, torque constant, wheel i

with $A \in \mathbb{R}^{9\times 9}$ and $B \in \mathbb{R}^{9\times 3}$ built from I, r_{cm}, and the per-wheel parameters $I_{w,i}, C_{w,i}, K_{m,i}$ (Table 10). The LQR design generalizes directly: K_d becomes a 3×9 gain matrix, using the same discretization, cost function, and offset filter as Table 9, with Q, R weighting all three tilt axes, body rates, and wheel speeds. The mode-transition logic gains a third mode, Edge-to-Corner jump-up, triggered once edge balancing is stable, followed by a second threshold switch into the corner LQR once estimated orientation nears corner-up.

5 Preliminary Hardware Design

This design is preliminary: final actuator and sensor specifications are not yet confirmed, so the mechanical and electrical architecture is kept modular and largely parametric (Section 1.1).

5.1 System Architecture

Figure 4 shows the closed control loop that ties the four subsystems together. The Teensy 4.1 runs estimation and control at 1 kHz and issues torque commands over CAN-FD to three moteus-n1 drivers, each closing a 30 kHz field-oriented current loop on its MN4006 motor. The resulting reaction torque acts on the cube body; the body's attitude is sensed by the BMI270 IMU and the rotor angle by the MA600 absolute encoder, closing the loop back to the controller. The two loops run at different rates by design: the fast current loop is delegated entirely to the drivers, so the flight controller need only meet the 1 ms outer-loop deadline of Section 4.4.

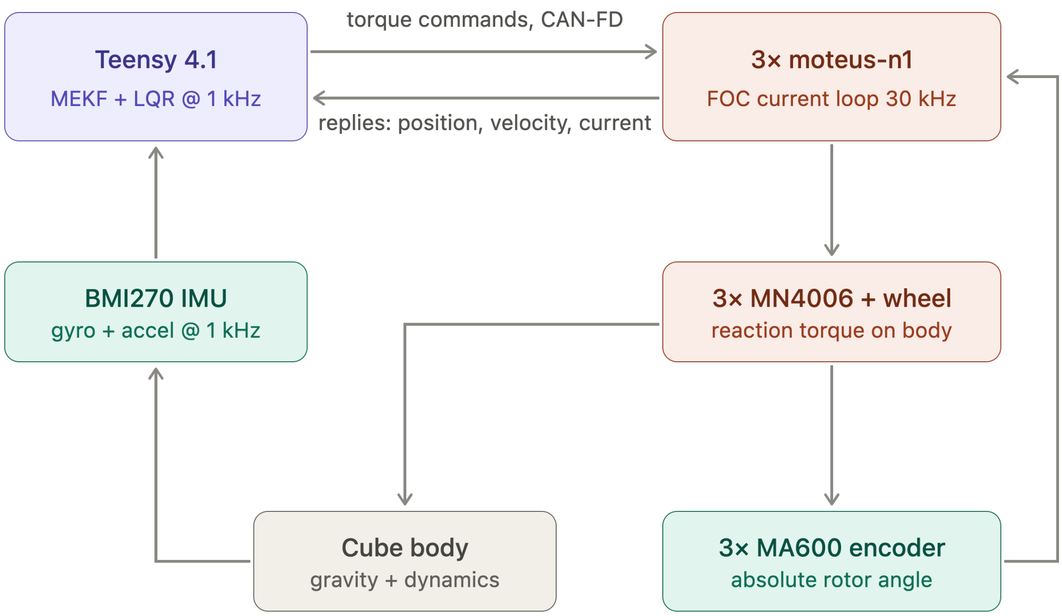

Figure 4: System architecture and control loop. Blue: processing; green: sensing; orange: actuation. Rates shown are the design targets of Section 4.

5.2 Sizing Constraints and Closure Procedure

The cube edge length L and the reaction-wheel inertia I_w are not independent design choices: they are the two ends of a single constraint chain that begins with what must physically be carried inside the cube. This section states that chain and the order in which it is closed. No dimensions are committed here—component selection is still open (Section 5), and the numbers follow from the procedure rather than preceding it.

The battery and the internal structure drive the envelope. The dominant volume claim inside the cube is not the reaction wheels but the power and support hardware. A LiPo pack sized for the jump-up duty cycle is a monolithic, incompressible block: unlike the electronics it cannot be redistributed into the corners or thinned, and its dimensions follow from the required capacity and discharge rating rather than from any packaging preference. Around it must sit an *inner* *structure*—a rigid subframe that locates the three orthogonal motors on mutually perpendicular axes through a common centre, carries the three motor drivers, the flight controller, the IMU and the brake servos, and reacts the motor mounting loads into the outer frame. The three wheel/motor modules must not intersect one another, the pack, or this subframe, and the pack should sit near the geometric centre so that it does not skew the cube's centre of mass away from the body diagonal.

The minimum internal clear volume is therefore set by the pack, the subframe and the electronics stack together, and L is the smallest edge length that contains that volume plus the wall thickness and the corner and edge contact features. This is the first quantity to fix, and everything downstream depends on it.

The envelope caps the wheel, and the wheel must meet the momentum requirement. Once L is set, the largest wheel outer diameter D_w that clears the subframe, the motor body and the adjacent orthogonal modules is a geometric consequence, and it caps the achievable inertia—not only through $I_w \sim m_w D_w^2/4$, but because any ballast necessarily sits inboard of the ring it displaces, so a wheel that is short of target at its maximum diameter cannot be rescued by adding mass. Against this stands the requirement from the jump-up analysis (Section 4.3.1),

$$I_{w,\mathrm{target}} = \frac{h_w(M, L)}{\omega_{\max}}, \qquad h_w \propto M L^{3/2},$$

which grows with both the cube mass and the edge length. Enlarging the cube to gain wheel diameter therefore also raises the requirement—but the achievable inertia grows faster than the requirement does (D_w^2 against $L^{3/2}$), so the loop does converge, and a wheel that is short at a given L can be closed by a modest increase in L.

Mass closes the loop. The cube mass M is not merely a logistics figure: it re-enters the dynamics through the gravitational tip-over torque $\tau_g = MgL/2$, which sets the floor the brake torque must exceed, and through $h_w \propto M$. Growing the envelope to fit a larger wheel adds frame and wheel mass, which raises $I_{w,\mathrm{target}}$ again. Mass, volume and inertia are thus one coupled system, closed by the iteration of Table 11 rather than by any of the three in isolation.

Table 11: Sizing closure procedure. Steps are executed in order; a failure at step 4 returns to step 3 or step 1 as indicated.

Step	Determines	Governing consideration
1	Battery pack geometry	Capacity and discharge rating for the jump-up duty cycle
2	Inner clear volume $V_{\min}$, hence minimum L	Pack + motor subframe + driver/controller stack, non-intersecting; pack near centre
3	Maximum wheel diameter D_w, hence achievable I_w	Clearance to subframe, motor body and the two orthogonal modules
4	Requirement $I_{w,\mathrm{target}} = h_w(M, L)/\omega_{\max}$	Corner jump-up, Eqs. (4)–(5)
5	Accept or iterate	If $I_w < I_{w,\mathrm{target}}$: raise L (step 2), raise $\omega_{\max}$, or trim M; re-enter at step 3
6	Mass reconciliation	Updated M feeds back into step 4; iterate to a fixed point

Inertia requirement is corner-critical. The two contact cases are not equally demanding. The corner equilibrium requires a larger rise of the centre of mass and a less favourable momentum-transfer geometry than the edge equilibrium, so for a common maximum wheel speed $\omega_{\max}$ it demands the greater wheel inertia (Table 7). Step 4 above is therefore always evaluated for the corner case; a wheel that satisfies the corner requirement satisfies the edge requirement with margin. Table 12 records both once the procedure closes.

Table 12: Required per-wheel inertia by contact case, to be evaluated at the selected $\omega_{\max}$ once the envelope is fixed.

Contact case	h_w (kg m^2/s)	$I_{w,\text{target}}$ (kg m^2)
Edge	TBD	TBD
Corner (critical)	0.198	3.15×10^{-4}

Mass budget. The allocation of Table 13 is the bookkeeping side of steps 5 and 6: it is populated as components are selected and re-totalled on each iteration, since the total is an input to $I_{w,\text{target}}$ and not merely an output of the design.

Table 13: Mass budget (per cube), to be populated as components are selected. The total is an input to the inertia requirement, not only a result.

Subsystem	Allocated mass (g)	Status
Battery pack	254	Drives envelope (step 1)
Reaction wheels (3×)	109 (3×)	Sized (Sec. 5.3.4)
Motors (3×)	68 (3×)	Drives reaction wheels
Inner structure / motor subframe	120	Transfers torque
Outer frame / contact features	350	Sized
Electronics (MCU, drivers, IMU, servos)	150	Allowance
Wiring / fasteners / margin	100	Allowance
Total M	1505	**Closed by iteration**

5.3 Mechanical Design and CAD Construction

This subsection states the mechanical concept and the CAD conventions the build follows. It is deliberately kept to the level of decisions and interfaces: the part-by-part model tree, the manufacturing drawings, the fabrication settings and the assembly sequence are recorded in Appendix B, so that a change to a bracket does not require an edit to the design rationale, and the rationale is not buried in geometry.

5.3.1 CAD Methodology and Model Conventions

The model is built top-down from the envelope closure of Section 5.2 rather than bottom-up from individual parts, so that the edge length L and the wheel diameter D_w propagate to every dependent feature when an iteration changes them. Three conventions make that propagation reliable, and they are stated here because they constrain how every part in Appendix B is modelled:

1. **A single skeleton drives the assembly.** L, D_w, the motor axis offsets and the wall thickness live in one parameter table referenced by every part; no downstream part carries a hard-coded dimension that duplicates one of them.
2. **The body frame is the CAD origin.** The assembly origin is the cube's geometric centre, with axes on the three motor spin axes, so that CAD-reported mass properties are directly the I and r_{cm} of Table 10 without a change of basis.

3. **Mass properties are an output of the model, not an annotation.** Every part carries its assigned material and print density, so the assembly mass roll-up is the quantity fed back into Table 13 at step 6 of Table 11.

Convention 3 is what makes the iteration auditable: the mass budget and the inertia requirement are read from the same model, so a design that closes on paper cannot silently fail to close in the CAD.

5.3.2 2D Prototype Build

The single-face rig is the first article built and the bench on which the parameters of Table 6 are measured. Its structure is a 3D-printed plate sized to one cube face, a central motor and wheel mount, a corner bearing defining the single rotational degree of freedom, and encoders and other hardware. Note that the servo (brake system) is not included in the 2D prototype as its purpose is validating the control loop on a more well-understood problem. Detailed geometry and the drawing set are in Appendix B.3.

5.3.3 3D Cube Frame

The frame is conceived in two parts, reflecting the constraint chain of Section 5.2. An *inner structure* locates the three motors on mutually perpendicular axes through a common centre and carries the battery pack, the three drivers, the flight controller, inertial measurement unit, and the brake servos; it is the element that fixes the minimum internal clear volume and therefore the edge length L. An *outer frame* carries the edge and corner contact features, protects the wheels, and reacts the motor mounting loads from the inner structure into the ground contact. Detailed geometry is pending CAD and follows the envelope closure.

The split is also the main mechanical interface in the machine, and it is managed as one: the inner structure is designed against a fixed bolt pattern and keep-out envelope published to the outer frame, so the two can be revised independently as long as that interface holds. The controlled interfaces are listed in Table 14; the parts realising them, with their drawings, are in Appendix B.4.

Table 14: Controlled mechanical interfaces. Each is frozen once agreed, so that the parts on either side can be revised independently.

Interface	Between	Controlled quantities
Motor mount	Motor $\leftrightarrow$ inner structure	Bolt circle, shaft height, concentricity to the wheel axis
Wheel hub	Wheel $\leftrightarrow$ motor rotor	Bore, bolt pattern, axial location, balance datum
Frame joint	Inner structure $\leftrightarrow$ outer frame	Bolt pattern, keep-out envelope, load path to the contact features
Electronics mount	Board $\leftrightarrow$ inner structure	Board outline, mounting-hole pattern, keep-out heights, connector exit directions (Sec. 5.4.5)
Brake mount	Servo $\leftrightarrow$ inner structure	Servo location, barrier throw
Battery retention	Pack $\leftrightarrow$ inner structure	Pack envelope, retention method, centring tolerance to the cube centre
Contact features	Outer frame $\leftrightarrow$ ground	Corner and edge geometry, material, replaceability

5.3.4 Reaction-Wheel Sizing

The wheel must supply the per-wheel inertia $I_{w,\text{target}}$ demanded by the corner jump-up (Section 4.3.1), at an outer diameter no greater than the envelope permits (step 3 of Table 11), while remaining light enough not to dominate the cube mass—since its own mass feeds back into $I_{w,\text{target}}$ through M. These three demands pull against each other, and the wheel architecture is chosen so that the design can be trimmed against them after fabrication rather than having to be right the first time.

Architecture: ballasted ring. Rather than adjusting inertia by scaling a solid disc, the wheel is built as a printed *ring plus spokes* whose inertia is then tuned by *ballast*: standard steel nuts seated in blind hexagonal pockets distributed around the ring. Each pocket both removes plastic and adds steel at the largest available radius, so the net inertia gain per pocket is large, and the target is reached by choosing how many bolts to install. This makes wheel inertia a discrete, reversible knob: the physical wheel can be re-tuned in either direction after fabrication to absorb the parameter uncertainty that the sizing procedure necessarily carries—most importantly the residual uncertainty in M, which is not known until the build is complete. A solid disc offers no such adjustment.

The design intent is that the bare printed ring supplies most of the required inertia and the ballast acts only as a trim, so that the count sits well inside its feasible range in both directions.

Parameters to be fixed. The geometry is defined by the quantities of Table **??**, all of which follow from the envelope closure of Section 5.2 and are recorded there once fixed.

Symbol	Parameter	Value
D_w	Ring outer diameter (capped by envelope)	13 cm
w_r	Ring radial width	1.5 cm
t_r	Ring axial thickness	1 cm
n_s, w_s	Spoke count and width	3, 1 cm
ρ_p	Printed-material density	$1290\,\text{kg/m}^3$
r_b	Ballast radius	5.75 cm
m_b	Net mass added per installed bolt	2.5 g
ΔI_b	Net inertia added per installed bolt	$8.27 \times 10^{-6}\,\text{kg}\,\text{m}^2$
N	Bolt count reaching $I_{w,\text{target}}$	15
m_w	Resulting wheel mass	109 g
I_w	Resulting wheel inertia	$3.4 \times 10^{-4}\,\text{kg}\,\text{m}^2$

The ballast is characterised *net* of the plastic each pocket displaces: m_b and ΔI_b are the increments the installed bolt-and-nut adds over the material it replaces, evaluated at the ballast radius. Sizing then reduces to solving

$$I_w(D_w, N) \;=\; I_{w,\text{ring}}(D_w) + N\,\Delta I_b \;\geq\; I_{w,\text{target}}$$

for the smallest admissible N at each candidate D_w, subject to the fit and mass checks below.

Sizing calculator. Rather than sweep a single geometric variable over a fine grid, we built a parametric calculator relating all of the variables in Table **??** to the final mass and moment of inertia of a single reaction wheel. There are limits on each variable, such as that the outer diameter of the reaction wheel D_w is within the body dimension L of the outer frame.

5.4 Electrical Design

The electrical architecture is presented here at the level of the four decisions that shape it: what the components are, how power is distributed, how the buses are routed and terminated, and how the whole is made safe to power on. The sheet-level schematics, the connector and pin tables, the harness drawings and the perfboard layout are collected in Appendix C, for the same reason the CAD detail is separated from the mechanical rationale — a connector re-pin should not require an edit here.

5.4.1 Components

The electronics and software will be built around a microcontroller communicating over a CAN bus with a motor driver that closes the current loop for the selected brushless motor. Wheel speed will be measured with a magnetic encoder on the motor shaft, and body tilt will be estimated from an onboard IMU. Braking will be actuated by a digital servo driven directly from the microcontroller, and a wireless module will provide a telemetry and debugging link. Power will be supplied by a LiPo battery, stepped down through a DC-DC converter for the logic-level electronics, with the motor driver powered directly from the battery bus.

The full parts list is catalogued in Appendix A (Table 26). The supplied set is an electronics-only BOM; the reaction wheels, the frame and all printed parts are the team's design responsibility, and their fabrication is the leading schedule risk.

Motor characterisation and operating envelope. Figure 5 shows measured torque and speed against phase current for the MN4006, taken from manufacturer bench data. The measurements were made with the motor driving propellers rather than reaction wheels, so the *load* is not representative of this application — an aerodynamic load rises with the square of speed, whereas a reaction wheel is an inertia with only bearing friction opposing it at constant speed. What does transfer is the motor's own electromechanical behaviour, since torque per ampere and back-EMF per unit speed are properties of the machine and not of what it is attached to. The curves are used here only in that sense: to bound the torque and current the actuator can be expected to deliver.

Three figures relevant to the design follow from the data. First, the torque–current relation in Figure 5(a) is close to linear ($r = 0.97$) with an incremental slope of 0.023 N m/A and a positive torque-axis intercept of about 0.08 N m. The intercept is the expected signature of losses that do not reach the shaft — magnetising and iron losses plus bearing friction — so the useful torque per ampere is the slope, and a constant inferred by simply dividing torque by current at a single point would overstate it (that ratio runs to 0.031 N m/A through the origin). Either way the motor produces at most 0.45 N m, an order of magnitude below the 5 N m brake torque assumed in Section 4.3.1. The two are not alternatives: the motor stores angular momentum in the wheel over about a second, and the brake destroys it in tens of milliseconds, so the impulsive torque comes from the collision and not from the drive. This is why braking is a separate mechanism rather than a motor function. Second, the highest measured point reaches 0.45 N m at 17.5 A, above the 16 A peak rating in Table 26, and that run terminated on the motor's thermal limit — so the top of this envelope is a short-duration capability, not a continuous one. Third, speeds cross the 6000 rpm design ceiling of Section 5.2 only under the lighter loads, which is consistent with that ceiling being set by bus voltage and derating rather than by any inability of the motor to spin faster.

5.4.2 Power Architecture

The power architecture is shown in Figure 6. A single 6S LiPo pack (22.2 V nominal, 3600 mA h) feeds everything through one XT90 connector, behind a combined E-stop and fuse using an

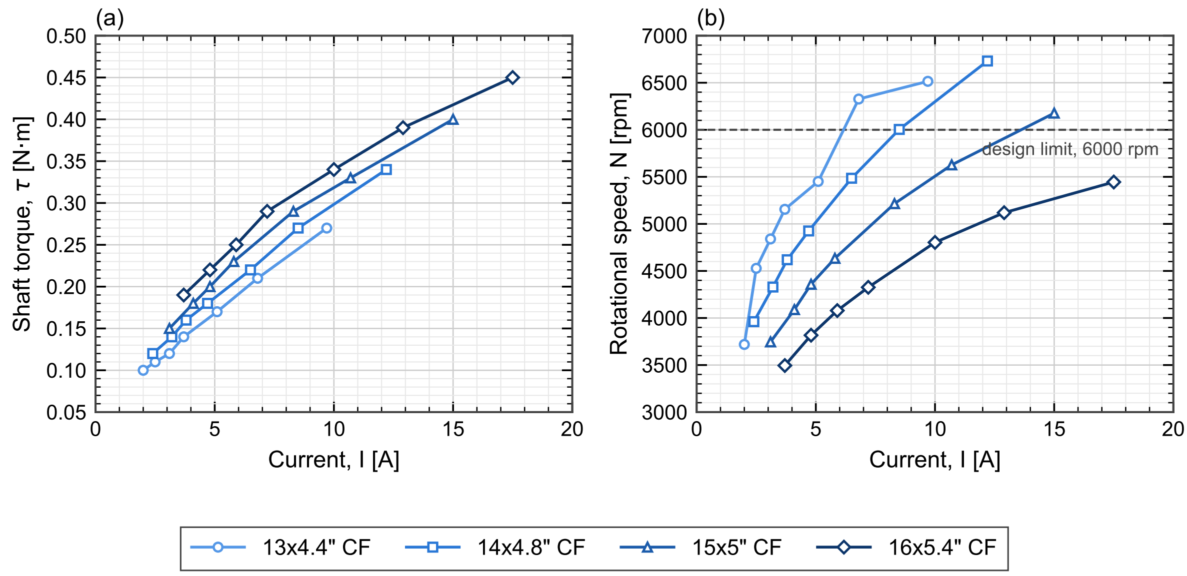

Figure 5: Measured MN4006 (a) shaft torque and (b) rotational speed against phase current, for one motor driving four different loads. Symbols are measured points at throttle settings of 50, 55, 60, 65, 75, 85 and 100 %; lines join successive points and are not fits. The dashed line in (b) is the 6000 rpm reaction-wheel design ceiling of Section 5.2. Loads shown are propellers, not reaction wheels: the curves characterise the *motor*, and the load-dependent spread between them does not carry over to this application. Manufacturer static bench data; the heaviest load reached the motor thermal limit at 100 % throttle.

XT90-S anti-spark connector to limit inrush into the driver bulk capacitance. The bus then splits into two rails. The *motor bus* runs at pack voltage (22–25 V across the discharge curve) directly into the three moteus drivers, with bulk capacitance rated to at least 50 V to absorb regenerative braking transients—which are not incidental here, since the jump-up maneuver of Section 4.3 dumps the full wheel kinetic energy back into the bus. The *logic rail* is derived by an LM2596 buck converter stepping 22 V down to 5 V for the Teensy, the ESP32 telemetry link, the brake servo and the IMU. Keeping the two rails separate downstream of the E-stop means motor-current ripple does not reach the sensing and processing electronics.

The rail-by-rail current budget that sizes the buck converter and the harness conductors is Table 15. It is populated from measurement during bench bring-up rather than from datasheet maxima alone, because the figure that matters for the 5 V rail is the coincident worst case — telemetry transmit burst, servo stall and full sensor load at once — and not the sum of typical values.

5.4.3 Schematic and Interconnect Diagrams

The signal-level design is documented as a set of sheets rather than a single drawing, so that each can be checked against the constraint that governs it. Table 16 is the index; the sheets themselves are in Appendix C.1, together with the connector and pin-assignment tables that a person holding the harness needs and a reader of this section does not.

5.4.4 Bus Topology, Termination and Signal Integrity

The CAN-FD bus is the one electrical subsystem whose physical routing is a correctness requirement rather than a packaging preference, and it is therefore fixed before the board outline is drawn. At the design data rate the bus must be a linear chain with short stubs and termination at exactly the two physical ends; a star topology or a mid-chain termination will pass a bench

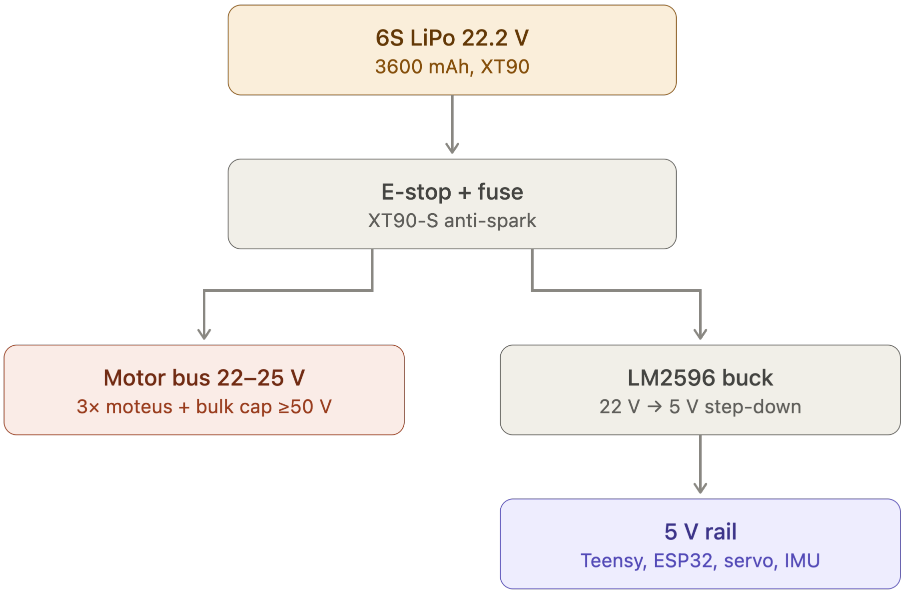

Figure 6: Power distribution tree. A single 6S pack feeds the motor bus directly and the 5 V logic rail through a buck converter, both behind a common E-stop and fuse.

continuity check and then fail intermittently under wheel vibration, which is the hardest class of fault to diagnose once the cube is closed. The design rules and their verification are listed in Table 17.

The encoder interfaces impose the complementary constraint. Each absolute encoder requires a short SPI run to its own driver and a tightly held air gap to its magnet, so the driver positions are dictated by the encoder positions, and those in turn by the motor axes. The physical layout order is therefore: motor axes fix the encoders, encoders fix the drivers, drivers fix the bus route, and only then is the board outline of Section 5.4.5 drawn around what remains.

5.4.5 Board Layout and Harness

The electronics are built on a double-sided perfboard rather than a fabricated PCB. This is a schedule decision, not a technical preference: fabrication turnaround does not fit inside the build window, and a perfboard can be modified in place when bring-up reveals a change. Its consequence is that layout discipline — rail separation, decoupling at each device, deliberate routing of the bus pair — must be enforced by hand and checked explicitly, since no design-rule check will do it.

The board outline, mounting-hole pattern, keep-out heights and connector exit directions constitute the electronics-mount interface of Table 14 and are frozen once delivered to the mechanical design: later electrical changes must fit inside that outline rather than move it. The harness — battery lead, driver drops, bus chain and servo drive — is fabricated to the drawings in Appendix C.3 and is labelled and strain-relieved before installation, because every connector in the machine sits on a structure that vibrates by design.

5.4.6 Electrical Safety and Bring-Up Protocol

Two rules govern all electrical bring-up and are stated here because they are design constraints, not merely lab practice. First, every first-time power application is made on a current-limited

Table 15: Rail current budget. Measured values replace estimates during bench bring-up; the 5 V total is checked against the LM2596 rating, which derates without a heatsink.

Load	Rail	Typical (mA)	Worst case (mA)	Basis
Teensy 4.1	5 V	TBD	TBD	Datasheet
XIAO ESP32-C6 telemetry	5 V	TBD	TBD	TX burst governs
BMI270 IMU	5 V	TBD	TBD	Datasheet
MG92B brake servo	5 V	TBD	TBD	Stall current governs
MA600 encoders (3×)	5 V	TBD	TBD	Datasheet
Logic rail total	5 V	**TBD**	**TBD**	**vs. converter rating**
moteus-n1 drivers (3×)	Pack	TBD	TBD	Spin-up, not balance, governs

Table 16: Schematic sheet index. Each sheet is checked against the governing constraint stated in its row before release.

Sheet	Content	Governing constraint to verify
1	Power entry and protection	E-stop, fuse rating, anti-spark inrush into the driver bulk capacitance
2	5 V logic rail	Converter rating vs. Table 15 worst case; back-feed diode; rail bulk capacitance voltage rating
3	CAN-FD bus	Linear topology, stub length, split termination at the two physical ends only (Sec. 5.4.4)
4	Encoder interfaces (3×)	SPI cable length and encoder air gap; driver placement follows from this, not the reverse
5	Flight computer and IMU	Sensor orientation and mounting location relative to the pivot (Sec. 4.4)
6	Telemetry link	Transmit-burst current and its return path off the logic rail
7	Brake servo drive	Stall current headroom on the logic rail; PWM routing away from the encoder SPI

bench supply, never on the battery; the battery is introduced only after the assembly has passed a full bench power-on. Second, the pack is never connected without the E-stop and fuse in circuit and the anti-spark path intact, since the driver bulk capacitance draws an inrush that will weld a bare connector.

The staged sequence — rails, then buses, then sensors, then open-loop motion, then closed-loop — is part of the validation plan and is given in Section 6. The isolation checks that precede first power-on (rail-to-rail separation, bus-to-chassis, and confirmation that no low-voltage capacitor sits on the pack bus) are recorded in Appendix C as a checklist, because they are performed on every re-assembly and not only once.

5.5 System Integration

5.5.1 Onboard Software Pipeline

The subsystems of Figure 4 are coordinated by a single deterministic task on the Teensy, whose structure is given in Figure 7. Each 1 ms cycle executes the same fixed sequence: sensor ingest reads the BMI270 over SPI and the moteus replies over CAN; the MEKF propagates attitude on the gyro and corrects it against the accelerometer while estimating gyro bias; the state

Table 17: Signal-integrity design rules and how each is verified. These are checked before cube closure, since all are hard to access afterwards.

Interface	Rule	Verification
CAN-FD bus	Linear chain; short stubs; split termination at the two physical ends only	Differential-pair scope capture; sustained error-frame count at full rate with all nodes addressed
Encoder SPI	Short run to its own driver; air gap and chip centring per the assembly spec	Read-back with no error flags; re-checked after mechanical assembly
IMU	Mounted at or near the pivot; orientation per the body frame of Sec. 5.3.1	Static orientation check; noise floor with wheels spinning
Rail separation	Motor bus and logic rail separated downstream of the E-stop	Ripple measurement on the logic rail under motor load
Grounding	Single return topology; no motor return current through the sensing ground	Continuity and isolation check before first power-on

vector $x = [g_B, \omega, h]$ is assembled from the estimator output and the wheel speeds; the LQR law of Section 4.4 evaluates $u = -Kx$ with a penalty on stored momentum h to bleed off wheel saturation; and the resulting commands leave as CAN torque requests to the drivers and a PWM signal to the brake servo. Telemetry is tapped off the assembled state to SD and Wi-Fi outside the control path, so logging cannot perturb loop timing.

The fixed ordering matters: with no dynamic allocation and no blocking calls, the worst-case cycle time is bounded by construction rather than measured after the fact, which is what makes the 1 kHz sample period of the discrete LQR design a design guarantee rather than an average.

5.5.2 Control Software Architecture

Section 4 derives the control law; this subsection states how that law is realised as software, which is a separate set of decisions. The implementation detail — the mode state machine, the estimator and controller in executable pseudocode, the fixed gains and tuning parameters, the timing budget and the fault responses — is collected in Appendix **??**, so that a re-tuned gain does not require an edit to the design sections.

The software is organised in the four layers of Table 18. The division is not stylistic: each layer has a different failure mode and a different verification method, and the interfaces between them are what allow the controller validated in simulation to be the same code that runs on the cube.

Table 18: Control software layers. Each is verified by a different method, which is why the boundaries are held.

Layer	Responsibility	Verified by
Hardware abstraction	Sensor and driver I/O, timing, bus transactions	Bench read-back; loop-timing measurement
Estimation	Attitude and rate estimate from IMU and encoders (Sec. 4.4)	Static and dynamic bench tests against a known reference
Control	State feedback, momentum management, saturation handling	Simulation on the identified model, then hardware
Supervision	Mode transitions, arming, watchdog and fault response	Fault-injection tests on the rig

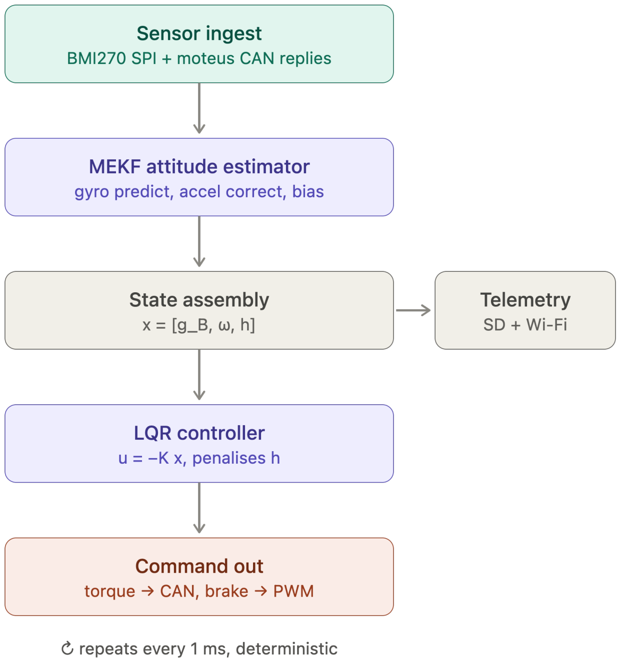

Figure 7: Teensy 4.1 control-loop software pipeline. The estimator and controller stages implement the design of Section 4.4; telemetry is a side-branch off the control path.

Operating modes. The machine is never in a single control regime. The supervisory layer sequences the modes of Table 19, with hysteresis on every threshold to prevent chattering at a boundary, and with a disarm transition available from every mode. Documenting the modes as a state machine rather than as a set of conditionals matters here because the dangerous transitions are the ones back down the ladder: a fall detected during corner balance must reach a safe state with the wheels commanded down, not merely switch to a different gain.

Real-time guarantees. The loop-rate claim of Section 4.4 is a design guarantee only if the worst-case cycle is bounded by construction, which is why the pipeline of Figure 7 uses a fixed stage ordering with no dynamic allocation and no blocking calls, and why telemetry is a side-branch rather than a stage. The per-stage timing budget and its measured margin are recorded in Appendix **??** and re-measured whenever a stage changes.

Fault handling. The failure cases the software must survive are sensor dropout, bus error, wheel saturation and loss of the balancing equilibrium. The first three have defined responses that keep the machine controlled; the fourth is a controlled shutdown rather than a recovery attempt, since a cube that has left its capture region cannot be recovered by the linear law and continued actuation only adds energy to the fall.

Table 19: Operating modes and transitions. Every mode has a disarm exit; the detailed conditions and hysteresis bands are in Appendix **??**.

Mode	Behaviour	Exit condition
Idle / disarmed	Drivers de-energised; sensors and telemetry live	Explicit arm command
Wheel spin-up	Open-loop speed ramp to the jump-up momentum target	Target wheel speed reached
Jump-up	Brake actuation and momentum transfer (Sec. 4.3)	Tilt inside the balancing capture band
Edge balance	Single-axis state feedback (Sec. 4.4)	Corner-transition command, or fall detected
Corner balance	Three-axis state feedback (Sec. 4.5)	Slew command, or fall detected
Slew / rotation	Reference tracking between equilibria	Reference reached; revert to balance
Fault / safe	Wheels commanded down, drivers disarmed	Manual reset

6 Testing and Validation Plan

This plan covers system-level validation. Parameter-identification tests that feed the model directly (swing test, current-step identification) are described with the model they validate in Section 4.2.

The plan is written before the design point is closed. Its methods, metrics and orderings are therefore fixed here, while the numeric thresholds that depend on the cube mass, edge length, brake torque or wheel geometry are carried as TBD and populated by the closure procedure of Section 5.2. This asymmetry is deliberate: a validation plan whose thresholds are pending is a plan awaiting data, whereas one whose methods are pending is not yet a plan. Where a threshold is already fixed elsewhere in this document — a gate criterion of Table 3, or a device limit from Appendix A — it is cited here rather than restated as a new requirement.

6.1 Validation Strategy and Stage Gates

Validation proceeds in ten stages, ordered so that each one removes the uncertainty that would otherwise invalidate the next. Three orderings in particular are not matters of convenience. Feasibility is verified before fabrication (V1 before V4), because a wheel that cannot meet its inertia target is cheaper to discover in a spreadsheet than in filament. The control law is exercised in simulation before it is exercised on hardware (V3 before V7), because a divergent gain set is diagnosed safely in simulation and destructively on a rig. Plant identification precedes control validation (V6 before V7), because gains derived from assumed parameters are not the gains that will be flown.

The governing rule is that no stage is entered until the exit criterion of the preceding stage is met. Where a stage fails, the response is to return to the stage that produced the offending assumption rather than to proceed with a known-invalid input; Section 2.3 records the contingency for the schedule consequences of doing so.

Gate G6 is a hard kill for the jump-up objective: if edge balance is not demonstrated by 13 August, V10 is abandoned rather than attempted, since a cube that cannot hold an edge cannot re-stabilise after arriving on one. This is a consequence of the objective ordering of Section 1.3, in which jump-up is a capability built on balance rather than an alternative to it.

6.2 Pre-Fabrication Feasibility Verification

The purpose of this stage is to establish, before any part is committed to fabrication, that the wheel about to be built can deliver the jump-up manoeuvre. This is a stricter question than whether the wheel meets its inertia target, because the target is itself the output of a model whose assumptions are not exact.

6.2.1 Two Independent Routes, Compared

Section 4.3 contains two derivations of the jump-up requirement that proceed from different assumptions, and this stage evaluates both rather than selecting one. The collision relation of Section 4.3 treats the wheel–body interaction as a perfectly inelastic impact and returns the required wheel speed ω_w directly from the body inertia and mass distribution. The torque-limited momentum budget of Equations (4) and (5) instead treats the achievable wheel speed as fixed and returns the required inertia $I_{w,\text{target}}$, while additionally accounting for a finite brake torque that the collision relation idealises away.

Agreement between the two is the evidence that the sizing is sound; disagreement is a finding in its own right and is reported rather than reconciled by choosing the more favourable result. The comparison is recorded in Table 21.

Table 20: Validation stages. Exit gates refer to Table 3; no gate or date is introduced here that is not already defined there.

#	Stage	Entry condition	Pass criterion	Exit gate
V1	Pre-fabrication feasibility (Sec. 6.2)	Candidate design point from Sec. 5.2	Both jump-up routes agree; margin held	G2
V2	CAD and dimensional verification (Sec. 6.3)	V1 passed; model complete	No interference; mass roll-up within budget	G2
V3	Simulation-first control (Sec. 6.4)	2D model of Sec. 4.4	Stable under all test cases and perturbation	G2
V4	Fabrication and mechanical acceptance (Sec. 6.3)	V1, V2 passed	Parts within dimensional tolerance; wheel balanced	pre-G3
V5	Electrical bring-up (Sec. 6.5)	V4 parts available	All criteria of Table 22	G3, G5
V6	1-DoF plant identification (Sec. 6.7)	V5 single axis live	Four parameters measured to tolerance	G3
V7	1-DoF control validation (Sec. 6.8)	V3, V6 passed	Rig balances $\geq$60 s unassisted	G3
V8	Edge balance (Sec. 6.9.1)	V7 passed; cube integrated	$\geq$60 s with disturbance recovery	**G6**
V9	Corner balance and slew (Sec. 6.9.2)	V8 passed	Point-contact regulation; slew tracked	G7
V10	Jump-up (Sec. 6.9.3)	V8 passed (hard kill)	Face-to-edge transition captured	G8

6.2.2 Brake-Torque Sensitivity

The brake-amplification factor $\beta = \tau_b/(\tau_b - \tau_g)$ of Equation (5) diverges as $\tau_b \to \tau_g$, so the inertia target is not merely uncertain in τ_b but unboundedly sensitive to it near the floor. A single assumed value of τ_b would therefore carry an unstated and potentially unbounded error into every downstream number.

This stage consequently requires τ_b to be reported as a sensitivity sweep rather than as a point value: $I_{w,\text{target}}$ is tabulated across a range of candidate brake torques, and the achieved wheel inertia is checked against each. The output of interest is the lowest brake torque at which the design still closes, which converts an unmeasured parameter into a stated requirement on the brake. That requirement is discharged when the brake torque is measured on hardware in Section 6.5; until then the sweep, not an assumption, is what the design rests on.

6.2.3 Trim Authority

Because $h_w \propto ML^{3/2}$, a mass overrun raises the inertia requirement rather than leaving it fixed, and the wheel must therefore retain the ability to be corrected in both directions after the cube is weighed. The ballast scheme of Section 5.3.4 provides this as a discrete trim, and this stage records how much of it remains unspent at the selected operating point: a design sitting only marginally above its target has no authority to absorb a mass overrun and is recorded as failing this gate even when the nominal inertia check passes. The trim is reported as the number of ballast elements that may be added or removed before the target is missed.

Table 21: Pre-fabrication feasibility gate. Pass conditions are stated as relations because the quantities they relate are outputs of the closure procedure of Section 5.2, not inputs fixed here.

Quantity	Source	Value	Pass condition
Required wheel speed ω_w	Collision relation, Sec. 4.3	TBD	$\leq$ achievable speed with margin
Achievable wheel speed ω_{max}	Drive analysis, Sec. 4.3.1	TBD	Sustained in service, not no-load
Speed margin $\omega_w/\omega_{\text{max}}$	Derived	TBD	< 1 by a stated margin
Target inertia $I_{w,\text{target}}$	Eq. (5)	TBD	—
Achieved inertia I_w	Wheel design, Sec. 5.3.4	TBD	$\geq I_{w,\text{target}}$
Route disagreement	Both of the above	TBD	Reported; explained if large
Ballast trim authority	Sec. 5.3.4	TBD	Non-zero in both directions
Nut retention margin	Sec. 5.3.4	TBD	Retention load carried by a stated feature

6.3 CAD and Dimensional Verification

The checks below split into those that run against the model, before anything is fabricated, and those that run against the physical part afterwards. The driving dimensions themselves are outputs of the packaging closure of Section 5.2 and are not verified here; what is verified is that the model realises them consistently and that the fabricated part matches the model.

Model checks (pre-fabrication). Interference and clearance verification on the wheel-to-subframe and inter-module clearances c_w and c_o of Table 27, through a full rotation of each wheel and across all three axes simultaneously; orthogonality of the three motor axes; position of the centre of mass relative to the geometric centre, since the modelling of Section 4 assumes a known offset; and the mass roll-up against Table 13, which is the input to the feasibility gate of Section 6.2 and must be re-run whenever it changes.

Part checks (post-fabrication). Dimensional acceptance of the ballast pockets and their retention floor, since both are structural features rather than cosmetic ones and a print that is dimensionally out of tolerance at the pocket is a wheel that cannot be trusted at speed; measured wheel mass against the CAD prediction; and static balance of the assembled wheel, which is the first opportunity to detect a ballast placement error before it appears as vibration at speed.

The modelled-versus-measured comparison is recorded in Table 31 of Appendix B.6 and is not duplicated here. A discrepancy found at that step is carried back into $I_{w,\text{target}}$ and absorbed by ballast trim where the authority allows, per Section 6.2; where it does not, the finding returns to the closure procedure rather than being accepted.

6.4 Simulation-First Validation

The control law is validated in simulation on the 2D model of Section 4.4 before any closed-loop hardware run. The plant is constructed from the parameters of Table 6, which are estimates until Section 6.7 measures them.

Test cases. Three cases are run against the linear-quadratic regulator baseline: recovery from an initial tilt offset, which establishes the region of attraction; rejection of an impulse disturbance applied at the body, which is the simulated counterpart of the disturbance-recovery criterion at G6; and behaviour under wheel-speed saturation, which exercises the momentum-dumping path of Section 5.5.2 and is the case most likely to be missed by a purely linear analysis, since saturation is precisely where the linear model ceases to describe the plant.

Robustness. Each case is re-run with the plant parameters perturbed about their nominal values, by a magnitude TBD set once the identification tolerances of Section 6.7 are known. The purpose is to establish that the gains are not knife-edge: a gain set that stabilises only the nominal plant is not a result, because the nominal plant is not the one that will be built.

Direction of validation. Simulation predicts and hardware corrects. The measured parameters of Section 6.7 are substituted into the plant and the gains re-derived before hardware balance is attempted; the simulation is validated against the measured plant, never the reverse. A discrepancy between simulated and measured closed-loop behaviour is treated as a finding about the model, and Section 6.10 defines the metrics in which that comparison is expressed.

6.5 Electrical Bring-up and Isolation

Bring-up is staged so that each power step is taken only once the preceding one has been shown safe: a current-limited bench supply before the battery, wheels removed before wheels fitted, and a single axis before three. The rationale is that each step removes one class of fault while the energy available to it is still bounded — the first battery connection is not an occasion to discover a wiring error. The detailed bring-up and isolation checklists are given in Appendix C and are not restated here; what follows is the set of numeric criteria that close each step.

The brake torque τ_b assumed in the sensitivity sweep of Section 6.2 is measured during this stage. Because the sweep establishes the lowest brake torque at which the design closes, the measurement is a pass/fail against that figure rather than an open-ended characterisation.

6.6 Sensor Testing

Section 1.1 identifies sensing, rather than the control law, as the effect that sets the achievable balancing limit in practical replications of this system. This subsection is the evidence for that claim, and its results bound what the control performance of Section 6.10 can be expected to achieve.

Inertial measurement. Static Allan deviation of the gyroscope over a logging run long enough to resolve the bias-instability floor, which gives the drift rate the attitude estimate must be corrected against; accelerometer noise floor in the same conditions; and the resulting bound on the tilt estimate as a function of the complementary-filter cross-over frequency.

Encoder. Air-gap verification to the recorded assembly spec, repeated after mechanical bolt-up, since the gap can shift when the structure is assembled — this is why the re-check appears as a separate step in Table 30 rather than being folded into the initial fitting. Linearity is checked over a full revolution against a known reference, and error flags are monitored over a sustained spin rather than sampled once.

Fusion. The complementary filter of Section 4.4 is validated against a known static tilt and a known dynamic profile, and the cross-over frequency is selected from the measured noise and drift figures above rather than assumed. The estimator's behaviour under the accelerometer trust gating of Appendix **??** is exercised deliberately, by applying a body acceleration large enough to

Table 22: Electrical acceptance criteria. Task references are to the electrical workstream schedule, which is the source of each figure.

Item	Criterion	Task
Motor incoming inspection	Phase-to-phase resistance within ±20% of the 194 mΩ datasheet figure on all three pairs; driver enumerates at a unique CAN ID	EL-13
Encoder air gap	≤0.5 mm, recorded as a numeric shim value, chip centred on the magnet; monotonic absolute angle over a full revolution with zero error flags	EL-14
Logic rail	5.00 V ±0.05 V at no load; ≥4.75 V at 1.5 A; ripple and buck case temperature recorded	EL-15
CAN bus, rig	Unpowered CANH–CANL 60 Ω ±3 Ω; all nodes enumerate at 5 Mbit/s	EL-16
CAN bus, integrated	60 Ω ±3 Ω with termination at the two physical ends only; zero error frames over a 30-minute soak with all three drivers commanded	EL-24
Bus transient	Bus holds ≥20 V at the three-axis current transient, measured at the board rather than the battery terminals; no fuse opening, no driver undervoltage fault	EL-25
Telemetry rail disturbance	Logic rail sag <100 mV during a 300 mA transmit burst; packet loss stated as a percentage	EL-22
Brake servo	Full commanded range on 50 Hz PWM; logic rail disturbance <100 mV during a servo step; flyback path verified	EL-31
Integrated power-on	All three axes addressable on battery power, all encoders within the gap spec after assembly, zero CAN error frames during a three-axis open-loop spin	EL-27

trip the gate, since an estimator whose gating is never tested is one whose gating is not known to work.

6.7 1-DoF Plant Identification

Four parameters are measured on the single-axis rig and returned to the control design of Section 6.4. Each is measured by a method chosen so that the quantity of interest dominates the measurement rather than appearing as a small correction to something else.

Tolerances are stated with the measured values, since a parameter without a tolerance cannot support the robustness sweep of Section 6.4 that consumes it. Loop latency is measured through the complete path rather than estimated from the 1 kHz loop rate, because the scheduling and bus-transport contributions are the ones a rate figure omits.

6.8 1-DoF Control Validation

The single-axis rig is the protected experimental item of Section 2.3, and its balance result is the evidence carried into the document. Gate G3 requires the rig to balance for at least 60 s unassisted.

Beyond the duration criterion, three quantities are recorded because they determine whether the result transfers to the integrated cube rather than merely holding on the rig. Recovery from a defined impulse establishes that the balance is a stabilised equilibrium and not a fortunate initial

Table 23: Plant parameters measured on the 1-DoF rig. Expected values follow from the design point and are populated when it closes.

Parameter	Method	Instrument	Value
Torque constant K_t	Current step against a locked rotor; torque from measured reaction	Driver current telemetry	TBD
Body inertia J	Free-swing test about the pivot; inertia from the measured oscillation period	Encoder or IMU log	TBD
Friction	Wheel coast-down from speed; viscous and Coulomb terms separated by fit	Encoder log	TBD
Loop latency	Command-to-response timestamp through the full sense–compute–actuate path	Loopback on the flight computer	TBD

condition. Steady-state wheel-speed drift measures momentum accumulation: a controller that balances while monotonically spinning up is one that will saturate, and the drift rate predicts when. Control effort is compared against the continuous and peak current limits of the driver given in Appendix A, since a balance sustained only by exceeding the continuous rating is not a balance that survives to the demonstration.

6.9 Edge, Corner and Jump-Up Validation

The three hardware milestones follow the objective ordering of Section 1.3, each entered only on the passing of the previous.

6.9.1 Edge Balance

Regulation about a line-contact equilibrium, and the hard-kill gate for the jump-up objective. G6 requires balance for at least 60 s with disturbance recovery. Recorded: sustained duration; the maximum initial tilt from which the controller recovers, which is the experimental region of attraction and is compared against the simulated figure from Section 6.4; the disturbance-rejection bandwidth; and wheel-speed drift as in Section 6.8. Numeric targets beyond the gate criterion are TBD.

6.9.2 Corner Balance and Commanded Slew

Regulation about a point-contact equilibrium on all three axes simultaneously, which is the case the wheel was sized against and therefore the first test in which the sizing assumption of Section 6.2 is exercised directly. Recorded: sustained duration, per-axis control effort, and cross-axis coupling, the last being absent from the 2D model and observable only here.

Commanded rotation between equilibria is verified in the same stage: the cube is commanded through a slew and the tracking error against the reference recorded, together with the settling behaviour on arrival. Targets TBD.

6.9.3 Jump-Up

The face-to-edge transition, captured on video at a frame rate sufficient to resolve the manoeuvre, per G8.

This test closes the loop opened in Section 6.2. The wheel speed at brake release is measured and compared against the value predicted before fabrication, and the two routes of Section 6.2 are re-evaluated against the measured brake torque. The discrepancy is the quantitative measure of the inelastic-collision idealisation, and its measurement is the experimental refinement of ω_w that

Section 4.3 anticipates. Recorded: wheel speed at release, achieved versus predicted; whether the edge is reached and held; and, on a successful transition, the time to re-stabilise, which is the quantity that determines whether chained jump-up — the walking objective — is reachable.

6.10 Control Performance Evaluation

The metrics used above are defined once here so that simulated and measured results are expressed in the same terms and remain comparable across stages.

Table 24: Control performance metrics. Definitions and methods are fixed; targets follow from the closed design point.

Metric	Definition	Measurement method	Target
Recovery angle	Largest initial tilt from which the equilibrium is regained	Ramped initial-condition release	TBD
Settling time	Time to enter and remain within a stated band about the equilibrium	Step or impulse response log	TBD
Disturbance-rejection bandwidth	Frequency to which body disturbances are attenuated	Swept or impulsive disturbance	TBD
Slew tracking error	Peak and RMS deviation from the commanded trajectory	Reference versus estimated attitude	TBD
Control effort	RMS and peak wheel current over the run	Driver current telemetry	Within Appendix A limits
Momentum drift	Rate of change of mean wheel speed while regulating	Encoder log over a sustained run	TBD

6.11 Milestone Acceptance Criteria

Table 25 states the condition under which each objective of Table 4 is accepted as demonstrated, and the subsection that produces the evidence. It is the closing element of the traceability chain that runs from the objectives of Section 1.3, through the required capabilities of Section 3.3, to the test that confirms them.

Table 25: Objective acceptance criteria and the evidence that supports each.

Objective	Acceptance condition	Evidence	Gate
Edge balance	Balanced $\geq$60 s unassisted with recovery from a defined disturbance	Sec. 6.9.1	G6
Corner balance	Point-contact regulation sustained on all three axes	Sec. 6.9.2	G7
Controlled rotation	Commanded slew completed within the stated tracking error	Sec. 6.9.2	G7
Jump-up	Face-to-edge transition achieved and captured on video	Sec. 6.9.3	G8
Walking (stretch)	Chained jump-up with re-stabilisation between transitions	Sec. 6.9.3	—

7 Future Work

7.1 NMPC & Warm-Start Control (Future Implementation)

7.2 Autonomous Stand-up & Disturbance Rejection

7.3 AI-Driven Control & Dynamic Coefficient Tuning

7.3.1 Motivation: Why a Fixed Gain Is a Fixed Assumption

The balancing controller of Section 4.4 computes a single gain K_d offline, from an (A, B) pair built from the parameters of Tables 5 and 10. That gain is optimal for exactly one plant: the one those numbers describe. It is retained here as the baseline precisely because it is simple and verifiable, but its optimality is conditional in three ways that the hardware of this project is expected to violate.

- **Parametric drift.** C_b and C_w are not obtained from CAD and are carried as estimates until the identification tests of Section 6.7 (Table 6). Bearing friction, wiring drag and motor cogging also change with temperature, wear and reassembly, so the identified values are a snapshot rather than a constant. The inertia tensor I and the center-of-mass offset r_{cm} are subject to the same objection at the few-percent level, and r_{cm} in particular shifts whenever the pack or harness is disturbed.
- **Payload and configuration changes.** Any added mass changes m, r_{cm} and I simultaneously, and therefore changes A. A gain tuned for one configuration is not merely suboptimal for another; near a fully unstable equilibrium it may be destabilizing.
- **Contact and surface variation.** The pivot is modeled as an ideal frictionless point or line. A real corner resting on a real surface slips, deforms and dissipates energy in a way that is neither in the model nor constant across surfaces, and this is the dominant unmodeled effect for the walking objective of Section 1.3.

The standard remedy within the existing design is gain scheduling on $\hat{\theta}_b$ (Section 4.4), but scheduling interpolates between gains that were themselves computed from the same assumed parameters. It addresses the weakening of the small-angle linearization; it does not address a plant that differs from the model. The proposal here is to keep the verified feedback law and adapt its coefficients online.

7.3.2 Proposed Hybrid Architecture

The intended structure is deliberately conservative: a classical inner loop whose stability properties are understood, wrapped by a slow outer loop that adjusts the inner loop's coefficients. Writing ϕ for the tunable coefficient vector,

$$u[k] = -K_d(\phi[k])\,\hat{x}[k], \qquad \phi[k+1] = \phi[k] + \Delta\phi\big(\hat{x}[k],\, \hat{x}[k-1], \dots\big), \tag{17}$$

where $\hat{x}$ is the estimated state of Section 4.4 and $\Delta\phi$ is produced by the tuner. Two properties of (17) matter more than the choice of learning algorithm. First, the inner loop remains a linear state feedback at every instant, so the controller that runs on the hardware is the one already implemented and tested. Second, ϕ is updated on a timescale far slower than the control period Δt, which keeps the frozen-gain analysis approximately valid and prevents the tuner from acting as an unmodeled fast feedback path. Three candidate tuners are of interest, and they are not equivalent in cost:

- **Bayesian optimization** over a low-dimensional parameterization of Q and R, minimizing a measured cost from short balancing trials [**shahriari2016bayesopt**]. This is the cheapest option and the one with a direct precedent on this class of hardware: automatic LQR tuning has been demonstrated on an inverted pendulum with on the order of tens of experiments [**marco2016automaticlqr**], and a self-tuning LQR has been run on the Cubli's own ancestor platform at ETH [**trimpe2014learningcubli**]. It tunes between trials, not within them.

- **Adaptive dynamic programming**, which solves the LQR problem online from measured data rather than from (A, B), converging toward the optimal gain for the true plant without an explicit model [**lewis2012adp**]. This is the option with the strongest theoretical guarantees and the strictest excitation requirements.
- **Reinforcement learning**, in which a policy trained in simulation outputs $\Delta\phi$ continuously from the state-error history. This is the most capable and the least verifiable, and is treated below as the exploratory branch rather than the default.

A stated reservation. The literature supporting learned control is overwhelmingly validated in simulation or on well-instrumented hardware, and the failure mode of an online tuner on an open-loop-unstable plant is a fall, not a degraded score. Any deployment on this cube should therefore be gated behind: a projection of ϕ onto a set for which the frozen-gain closed loop is verified stable across the parameter uncertainty; a fallback to the fixed K_d of Section 4.4 on any estimator fault or divergence; and bench trials with the cube tethered. This is future work in the literal sense — none of it is a committed deliverable of the present design, and the baseline controller is not contingent on any of it.

7.3.3 Module 1: Dynamic Gain Tuning

The tuner does not emit motor currents. It emits the coefficients from which K_d is computed, which bounds what a bad update can do. Two parameterizations are worth comparing:

- **Weight-space tuning.** The policy outputs the diagonal entries of the cost matrices, $\phi = (\log q_1, \ldots, \log q_9,\ \log r_1, \log r_2, \log r_3)$ with $Q = \mathrm{diag}(q_i) \succ 0$ and $R = \mathrm{diag}(r_i) \succ 0$, and K_d follows from the discrete algebraic Riccati equation. The logarithmic parameterization enforces positive definiteness by construction, so every ϕ the policy can emit corresponds to a well-posed LQR problem and, for a stabilizable (A_d, B_d), to a stabilizing gain for the *modeled* plant. The cost is that a Riccati solve must run onboard whenever ϕ changes, which is what confines updates to a slow outer loop.
- **Gain-space tuning.** The policy perturbs the entries of K_d directly, or equivalently the K_p, K_d pair of a PD form, avoiding the Riccati solve at the price of losing the structural guarantee above. This is the practical choice if the outer loop must run fast, and it makes the projection step of the previous paragraph mandatory rather than merely prudent.

For the 3D corner-balancing case the state is $x = (\theta, \omega_b, \omega_w) \in \mathbb{R}^9$ and K_d is 3×9 (Section 4.5), so tuning the 54 free entries directly is not sensible on the available data. Weight-space tuning reduces the problem to 12 coefficients, and symmetry of the three wheel axes reduces it further if the wheels are treated as identical. A reward or cost of the form

$$c[k] = \hat{x}^{\mathrm{T}}[k]\, W_x\, \hat{x}[k] + \rho\, \|u[k]\|^2 + \sigma\, \|\omega_w[k]\|^2 \tag{18}$$

adds an explicit penalty on stored wheel momentum, which the nominal LQR cost $J(u)$ of Section 4.4 does not distinguish from any other state deviation. This term is the one directly relevant to the saturation problem raised in Section 1.2: it expresses a preference for holding attitude with the wheels near zero speed, leaving momentum headroom for disturbance rejection.

7.3.4 Module 2: Sim-to-Real Transfer via Domain Randomization

A policy trained against a single nominal parameter set learns to exploit that parameter set. Domain randomization instead samples the plant parameters from a distribution at the start of each training episode [**tobin2017domainrand**], so that the policy is optimized against a family of plants rather than one, and the transfer method has a substantial record on physical legged hardware [**hwangbo2019anymal**]. For this cube the randomized quantities follow directly from the identification gaps already documented:

- **Damping:** C_b, C_w over a wide interval, since these are estimated rather than measured until Section 6.7 closes.
- **Inertia and mass distribution:** I, m and r_{cm} perturbed by a few percent, covering print-density variation, fastener placement and harness routing.
- **Actuation:** K_m within motor tolerance, plus a current saturation limit and a commanded-torque delay, so the policy cannot assume an ideal actuator.
- **Contact:** pivot friction and restitution, and a small random ground slope — the term standing in for the surface variation that motivates this module.
- **Sensing:** gyro bias walk, accelerometer noise and an injected vibration component, matching the estimator degradation identified in Section 1.1 as the practical limit on balancing.

The last item deserves emphasis, because it is the one most often omitted. The accelerometer corruption on this platform is generated by the controller's own wheel activity, so it is correlated with the policy's actions rather than independent of them. A randomization scheme that injects only white sensor noise will understate the difficulty, and a policy that transfers under it should not be assumed to transfer on hardware. Whether the randomized distribution actually covers the real cube is an empirical question, settled by the same bench tests as everything else — the honest failure mode of this module is a distribution the designer chose to be convenient rather than representative.

7.3.5 Module 3: Jump-Up Energy Optimization

The jump-up analysis of Section 4.3.1 treats the maneuver as a spin-up to $\omega_{\max}$ followed by an abrupt stop, with all deviation from the ideal absorbed into a single brake-amplification factor $\beta = \tau_b/(\tau_b - \tau_g)$. That model is adequate for sizing and is deliberately conservative, but it is a poor description of what the mechanism does: the servo-driven barrier has finite travel, the collision is not instantaneous, and the resulting torque profile is neither a step nor known in advance.

The proposal is to treat the torque command over the maneuver window as a profile to be optimized rather than a constant. Parameterizing the pre-brake and post-impact torque commands by a small vector ψ — for instance the timing of brake release relative to the wheel-speed trajectory, and the shape of the recovery torque applied by the remaining wheels immediately after impact — the objective is

$$\min_{\psi} \; \mathbb{E}\Big[\|\theta(t_f) - \theta^\star\|^2 + \mu \, \|\omega_b(t_f)\|^2 \Big] \quad \text{subject to} \quad \omega_w \le \omega_{\max}, \;\; |u_i| \le u_{\max}, \tag{19}$$

where t_f is the instant the controller hands over to the balancing law at $|\hat{\theta}_b| < \theta_{\text{th}}$. Minimizing residual body rate at handover is the quantity of interest, because a jump that reaches the equilibrium with excess rate demands recovery authority the balancing controller may not have. Because each evaluation of (19) is a physical jump, the sample budget is small and the appropriate method is Bayesian optimization rather than a policy-gradient method, with the initial design taken from the analytical h_w of Equation (4).

Two constraints on this module must be stated. First, it presupposes the measurement described in Section 4.3.1: until the spin-down test of Section 6.9.3 reports both the impulse-equivalent mean $\bar{\tau}_b = h_w/\Delta t$ and the peak torque, there is no validated model to optimize against and no way to distinguish an improved profile from a mis-measured one. Second, the structural caveat carries over unchanged — optimization must not be permitted to search toward shorter stopping times, since τ_b is simultaneously a load on the barrier, bolt, bracket and spokes, and the sizing value was chosen for the dynamics and not for that load path. The search space must be bounded by the structural case, not by the dynamic one.

7.3.6 Relevance Beyond the Bench

Two application claims follow, and both are narrower than the general enthusiasm for learned control would suggest.

For **CubeSat attitude control**, the relevant fact is that a spacecraft's inertia tensor is known imprecisely at integration and changes in flight — propellant depletion, deployable appendages, thermal distortion — while reaction-wheel friction rises with bearing age. These are the orbital analogues of the parametric drift of Section 4.4, and they are conventionally handled by conservative fixed gains with margin, at a cost in pointing performance. A tuner that adapts Q and R against measured performance addresses exactly this gap. The qualification barrier for flying learned control is high and unaddressed here; the architecture of (17) is chosen partly because a bounded outer loop over a classical inner loop is the form most likely to survive that scrutiny.

For **low-gravity surface mobility**, the case is stronger, because the plant is genuinely unknown before arrival. Regolith mechanical properties on a small body are not characterized to the precision a hopping controller would want, and the flight-proven hoppers cited in Section 1.2 [5, 6] move ballistically rather than under closed-loop control for this reason. The contact randomization of Module 2 and the profile optimization of Module 3 map onto that problem directly: a hop whose landing attitude is controlled, on a surface whose properties were sampled from a distribution rather than assumed, is the capability that separates directional walking from a ballistic tumble [7]. The cube tests this under 1 g, where the destabilizing torque is larger and never relents — a harder case in that one respect, as argued in Section 1.2, though it does not reproduce the low-gravity contact dynamics that would decide the real maneuver.

7.4 Single-Wheel Variant / Extensions

References

[1] Mohanarajah Gajamohan, Michael Merz, Igor Thommen, and Raffaello D'Andrea. "The Cubli: A Cube that can Jump Up and Balance". In: *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. Vilamoura-Algarve, Portugal, Oct. 2012, pp. 3722–3727. DOI: 10.1109/IROS.2012.6385896.

[2] Mohanarajah Gajamohan, Michael Muehlebach, Thomas Widmer, and Raffaello D'Andrea. "The Cubli: A Reaction Wheel Based 3D Inverted Pendulum". In: *2013 European Control Conference (ECC)*. Zurich, Switzerland, July 2013, pp. 268–274.

[3] Michael Muehlebach and Raffaello D'Andrea. "Nonlinear Analysis and Control of a Reaction-Wheel-Based 3-D Inverted Pendulum". In: *IEEE Transactions on Control Systems Technology* 25.1 (Jan. 2017), pp. 235–246. DOI: 10.1109/TCST.2016.2549544.

[4] Ronny Votel and Doug Sinclair. "Comparison of Control Moment Gyros and Reaction Wheels for Small Earth-Observing Satellites". In: *26th Annual AIAA/USU Conference on Small Satellites*. 2012. URL: https://digitalcommons.usu.edu/smallsat/2012/all2012/74.

[5] Tetsuo Yoshimitsu, Takashi Kubota, and Ichiro Nakatani. "New Mobility System for Small Planetary Body Exploration". In: *Proceedings of the 1999 IEEE International Conference on Robotics and Automation (ICRA)*. 1999, pp. 1404–1409.

[6] Tra-Mi Ho, Vladimir Baturkin, Christian Grimm, Jan Thimo Grundmann, et al. "MASCOT—The Mobile Asteroid Surface Scout Onboard the Hayabusa2 Mission". In: *Space Science Reviews* 208 (2017), pp. 339–374. DOI: 10.1007/s11214-016-0251-6.

[7] Alexander Spiridonov et al. "SpaceHopper: A Small-Scale Legged Robot for Exploring Low-Gravity Celestial Bodies". In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. arXiv:2403.02831. 2024, pp. 3464–3470.

[8] Matthias Hofer, Michael Muehlebach, and Raffaello D'Andrea. "The One-Wheel Cubli: A 3D Inverted Pendulum That Can Balance with a Single Reaction Wheel". In: *Mechatronics* 91 (2023), p. 102965. DOI: 10.1016/j.mechatronics.2023.102965.

A Bill of Materials

The components supplied for this project constitute an *electronics* bill of materials: the actuation chain, wheel and attitude sensing, on-board compute, communications and power distribution are specified, but no structural or mechanical hardware is. The reaction wheels, the frame, the motor mounts, the fasteners and every printed bracket are the team's design responsibility, and are treated as such in Section 5. Table 26 lists the supplied items grouped by subsystem. The actuation figures derived from these parts — the torque constant, the achievable torque and wheel speed, and the inertia that sizes the wheel — are developed where they are used, in Sections 4.3.1 and 5.3.4.

The architecture the BOM implies is a two-level control stack: each mjbots moteus-n1 closes a high-rate field-oriented current loop on its motor, using the MA600 encoder for rotor angle, while the Teensy 4.1 runs the outer attitude-estimation and state-feedback loop over CAN-FD — the same inner-tachometer / outer-attitude layering used in spacecraft reaction-wheel assemblies. A mechanical wheel brake, following the original ETH Cubli [2], provides the near-impulsive momentum transfer required for the jump-up.

Procurement is the single largest schedule risk. The long-lead items are not electronic but mechanical: the frame stock and the printed structure gate the entire build, and with three actuation sets and no spare, the loss of a motor or driver in service would force the reduced-actuation one-wheel variant [8]. These items are ordered first. Consumables and bench equipment not covered by the supplied set — wire, connectors, fasteners, filament, the current-limited bench supply and the battery-safety kit — are tracked separately in the build documentation.

Table 26: Supplied bill of materials, grouped by subsystem.

Ref	Item	Qty[a]	Key specifications	Function
Actuation				
1	T-Motor MN4006 KV380	3	18N24P outrunner; 380 rpm/V; 68 g; ⌀44.35 mm × 21 mm; 194 mΩ; peak 16 A, 380 W	Reaction-wheel torque source
2	mjbots moteus-n1	3	10–54 V; 100 A peak / 9 A cont.; 15–30 kHz FOC; 5 Mbit/s CAN-FD; 14.6 g	FOC current/torque driver (inner loop)
11	TowerPro MG92B servo	1–3	metal-gear micro servo; ≈0.29 N m (≈3 kg cm) at 6 V; 50 Hz PWM	Wheel brake actuator (jump-up)
Sensing				
3	mjbots MA600 breakout + ring magnet	3	TMR absolute angle; SPI 6 MHz, 3.3 V; 16-bit (65,536 counts/rev); air-gap ≤ 0.5 mm	Wheel angle: commutation + state feedback
10	SparkFun BMI270 6-DoF IMU	1	3-axis accel + 3-axis gyro (no magnetometer); I^2C 400 kHz (Qwiic); ODR up to ≈ 1 kHz	Body attitude reference (tilt + rates)
Compute and Communications				
4	Teensy 4.1	1	600 MHz Cortex-M7 + FPU; native CAN-FD (CAN3); 5 V supply	Flight computer: estimator + control law
5	CAN-FD adapter (Teensy 4.1)	1	CAN-FD transceiver, logic-level → differential; 5 Mbit/s	Bus transceiver for the MCU

Continued on next page

Table 26 (continued)

Ref	Item	Qty[a]	Key specifications	Function
6	mjbots mjcanfd-usb-1x	1	USB ↔ CAN-FD, 5 Mbit/s	Bench interface (calibration, tuning)
7	JST-PH3 CAN cables	≥4	2 A AC/DC; 3-conductor daisy-chain	CAN bus interconnect
8	Seeed XIAO ESP32-C6 + antenna	1	Wi-Fi; 3.3 V UART to MCU; TX burst ≈ 300 mA	Wi-Fi telemetry / command link
9	60.4 Ω 0.5 W resistor	4	Split termination, 2 × 60.4 Ω ≈ 120 Ω bus impedance	CAN bus termination
17	4.7 nF capacitor	2	common-mode filter, bus midpoint to ground	CAN split-termination filter
Power				
12	Turnigy 6S LiPo (XT90)	1	22.2 V; 1550 mA h; 150C; ≈ 35 W h; ≈ 254 g	Primary energy store
13	LM2596 buck module	1	step-down to 5.0 V; rated 2–3 A (derates without heatsink)	5 V logic rail
14	XT90 connector pair	1–2	rated ≈ 90 A	Battery power connector
15	100 µF 16 V electrolytic	2–4	16 V part — 5 V rail only	5 V-rail bulk reservoir
19	1N5819 Schottky diode	1–2	1 A, 40 V	5 V-rail back-feed protection
Integration				
16	100 nF 50 V capacitor	6–10	local decoupling at each IC	Per-IC supply decoupling
18	Double-sided perfboard	1	15 cm × 9 cm, double-sided	Power / signal distribution board

[a] Quantity per assembled cube; motors, drivers and encoders are 3 each.

B CAD and Construction Detail

This appendix holds the mechanical detail deliberately kept out of Section 5.3: the model tree, the drawing set, the fabrication settings and the assembly sequence. The rationale for the architecture — why the frame is split, what fixes the envelope, how the wheel is tuned — stays in Section 5.3 and is not repeated here. Anything in this appendix may change without the design intent changing; anything in Section 5.3 may not.

B.1 Design Parameter Table

Table 27 is the single source of the driving dimensions referenced by every part in the model, per convention 1 of Section 5.3.1. It is the CAD-side mirror of the envelope closure of Section 5.2 and is updated on each iteration.

Table 27: Driving CAD parameters. Every dependent feature references these; no part carries a duplicated hard-coded value.

Symbol	Parameter	Value
L	Cube edge length (outer)	15 cm
t_w	Frame wall thickness	5 mm
D_w	Wheel outer diameter (Sec. 5.3.4)	13 cm
d_m	Motor axis offset from cube centre	TBD
h_m	Motor mounting face height on its axis	TBD
c_w	Wheel-to-subframe axial clearance	TBD
c_o	Clearance between orthogonal wheel modules	TBD
$V_{\min}$	Minimum internal clear volume	TBD
r_c	Corner contact feature radius	TBD

B.2 Model Tree and Part List

Table 28 lists the modelled parts by assembly, with their fabrication route and current status. It is the mechanical counterpart to the electronics bill of materials of Appendix A: the parts here are the team's design responsibility and are not supplied.

B.3 2D Prototype Construction

B.4 3D Cube Construction

B.5 Fabrication and Assembly

B.5.1 Print and Manufacturing Settings

The printed parts are structural, not cosmetic: the wheel ring carries centrifugal load at the ballast pockets and the motor mounts carry the brake impulse, which is the design load case identified in Section 1.1. Settings are therefore recorded per part in Table 29 and treated as part of the design, since a part reprinted at a different orientation or infill is not the part that was analysed.

B.5.2 Assembly Sequence

The sequence of Table 30 is ordered by access: each step is one whose verification becomes impossible or destructive once a later step is complete. The encoder air gap in particular is rechecked after mechanical assembly, since it can shift when the structure is bolted up (Table 17).

Table 28: CAD model tree and part list. Quantities are per assembled cube unless noted; the 1-DoF rig parts are listed separately.

Ref	Part	Qty	Fabrication route	Status
Inner structure				
M1	Motor subframe	TBD	TBD	TBD
M2	Motor mount (×axis)	3	TBD	TBD
M3	Battery cradle / retention	1	TBD	TBD
M4	Electronics tray	1	TBD	TBD
M5	Brake servo bracket	3	TBD	TBD
Reaction wheel assembly				
W1	Ballasted ring (Sec. 5.3.4)	3	TBD	TBD
W2	Wheel hub / rotor adapter	3	TBD	TBD
W3	Ballast bolt and nut set	TBD	Standard hardware	TBD
W4	Brake strike feature	3	TBD	TBD
Outer frame				
F1	Frame member / edge rail	TBD	TBD	TBD
F2	Corner contact feature	8	TBD	TBD
F3	Face panel / wheel guard	TBD	TBD	TBD
1-DoF rig (Sec. 5.3.2)				
R1	Face plate	1	TBD	TBD
R2	Pivot bearing mount	1	TBD	TBD
R3	Rig motor mount	1	TBD	TBD
R4	Hard stops / containment	TBD	TBD	TBD

B.6 Mass Properties Verification

The CAD roll-up is an estimate until the assembly is weighed. Step 8 above closes the loop of Section 5.2: the measured mass is compared against the modelled value, and any discrepancy is carried back into $I_{w,\text{target}}$ and absorbed, where possible, by the ballast trim of Section 5.3.4 rather than by a re-design.

Table 29: Fabrication settings for the structural printed parts. Layer orientation is specified relative to the principal load direction.

Ref	Material	Infill / walls	Orientation and rationale
W1	TBD	TBD	TBD
W2	TBD	TBD	TBD
M1	TBD	TBD	TBD
M2	TBD	TBD	TBD
F1	TBD	TBD	TBD
F2	TBD	TBD	TBD

Table 30: Assembly sequence. Ordering is set by access: each check must be completed before the step that encloses it.

Step	Operation	Check before proceeding
1	Wheel assembly: ring, hub, ballast bolts to the selected count	Balance check; bolt retention per Sec. 5.3.4
2	Motor to mount, encoder magnet to rotor	Encoder air gap and centring to the assembly spec
3	Motor modules to inner structure, three axes	Axis orthogonality; wheel clearance c_w, c_o
4	Battery and electronics tray into inner structure	Pack centring; board keep-out clearances
5	Harness install and dressing	Continuity and isolation (Sec. 5.4.6)
6	Inner structure into outer frame	Encoder air gaps re-checked after bolt-up
7	Contact features and guards	Contact geometry; wheel guard clearance
8	Mass properties measurement	Measured mass against Table 13

Table 31: Modelled versus measured mass properties. A discrepancy here is absorbed by ballast trim where the margin allows.

Quantity	CAD value	Measured	Discrepancy
Total cube mass M	TBD	TBD	TBD
Wheel mass m_w (each)	TBD	TBD	TBD
Wheel inertia I_w (each)	TBD	TBD	TBD
CoM offset from cube centre	TBD	TBD	TBD

C Electrical System Detail

This appendix holds the electrical detail kept out of Section 5.4: the schematic sheets indexed in Table 16, the connector and pin assignments, the harness drawings, the board layout and the bring-up checklists. The architectural decisions — two rails behind one E-stop, linear bus topology, perfboard rather than fabricated board — remain in Section 5.4. The part specifications are in Appendix A and are not repeated.

C.1 Schematic Sheets

C.1.1 Sheet 1 — Power Entry and Protection

C.1.2 Sheet 2 — Logic Rail

C.1.3 Sheet 3 — CAN-FD Bus

C.1.4 Sheet 4 — Encoder Interfaces

C.1.5 Sheet 5 — Flight Computer and IMU

C.1.6 Sheet 6 — Telemetry Link

C.1.7 Sheet 7 — Brake Servo Drive

C.2 Connector and Pin Assignments

Table 32 is the connector schedule. It exists so that a harness can be built and re-built from this document alone, and so that a mis-mate is a documented impossibility rather than a discovered one: no two connectors carrying different voltages share a housing type on this machine.

Table 32: Connector schedule. Keying and housing types are chosen so that two connectors carrying different voltages cannot be mated.

Ref	From	To	Housing / keying	Signals
J1	Battery pack	Protection / E-stop	TBD	Pack +, pack −
J2	Protection	Distribution board	TBD	Motor bus
J3	Distribution board	Driver, axis 1	TBD	Motor bus
J4	Distribution board	Driver, axis 2	TBD	Motor bus
J5	Distribution board	Driver, axis 3	TBD	Motor bus
J6	Board	CAN chain, end A	TBD	CAN-H, CAN-L, GND
J7	CAN chain, end B	Termination network	TBD	CAN-H, CAN-L, GND
J8	Driver, axis i	Encoder, axis i	TBD	SPI, 3.3 V, GND
J9	Board	IMU	TBD	Serial bus, supply, GND
J10	Board	Telemetry module	TBD	UART, supply, GND
J11	Board	Brake servo, axis i	TBD	PWM, 5 V, GND

C.3 Harness Drawings and Wire Schedule

Table 33 is the wire schedule. Conductor gauge on the motor bus is set by the spin-up current rather than the balancing current, since spin-up is the sustained high-current phase; the signal runs are set by length limits rather than by current.

Table 33: Wire schedule. Motor-bus gauge is sized on spin-up current; signal runs are length-limited (Table 17).

Ref	Run	Gauge	Length	Sizing basis
H1	Pack to protection	TBD	TBD	Peak bus current
H2	Protection to board	TBD	TBD	Peak bus current
H3	Board to driver drops (3)	TBD	TBD	Per-axis spin-up current
H4	Motor phase leads (3)	TBD	TBD	Motor phase current
H5	CAN chain segments	TBD	TBD	Topology, not current
H6	Encoder SPI runs (3)	TBD	TBD	Length limit governs
H7	Servo drives (3)	TBD	TBD	Stall current

C.4 Board Layout

The floorplan is constrained before it is aesthetic: the driver positions follow from the encoder cable limit, the bus route follows from the driver positions, and the board is placed around what remains (Section 5.4.4). Rail separation is maintained physically on the board, not only schematically, so that motor-current return does not share copper with the sensing ground.

C.5 Bring-Up and Isolation Checklists

These checks are performed on every re-assembly, not only on first build, since the encoder gaps and connector seating are exactly what mechanical work disturbs.

Before any power is applied.

1. Isolation: motor bus to chassis, logic rail to motor bus, each rail to ground.
2. Capacitor voltage ratings confirmed by position — no low-voltage part on the pack bus.
3. Polarity of every fabricated connector verified against Table 32.
4. Protection element in circuit: E-stop functional, fuse fitted and rated, anti-spark path intact.
5. Encoder air gaps re-measured after mechanical assembly.

First power-on, current-limited bench supply.

1. Current limit verified against a known load before connecting the assembly.
2. Logic rail voltage and ripple under load; converter thermal check.
3. All drivers enumerate on the bus; encoders read back with no error flags; IMU responds.
4. Bus integrity at the full data rate: sustained soak with all nodes addressed, error-frame count logged.
5. Open-loop wheel motion, wheels contained, one axis at a time.

First battery power-on. Performed only after the bench sequence passes in full, with the pack protection verified, cell monitoring active and the assembly in a fire-safe area. Inrush is measured against the bulk capacitance on connection.

C.6 As-Built Measurements

Table 34 records the measured electrical figures that replace the estimates used during design. These are the numbers reported back to the modelling and control work, and they are the reason the parameter identification of Section 4.2 runs on measured rather than assumed values.

Table 34: As-built electrical measurements. Each replaces an estimate used earlier in the design.

Quantity	Measured	Replaces / feeds
Motor torque constant K_m	TBD	Table 6, datasheet value
Wheel inertia I_w (spin-down)	TBD	Table 6, CAD value
Viscous and Coulomb friction	TBD	C_b, C_w estimates
Command-to-encoder latency	TBD	Loop timing, Sec. 5.5.2
Achievable wheel speed on bus	TBD	$\omega_{\max}$, Sec. 4.3.1
Logic rail load, worst case	TBD	Table 15
Runtime per charge under balance load	TBD	Demonstration planning